Review of last lecture

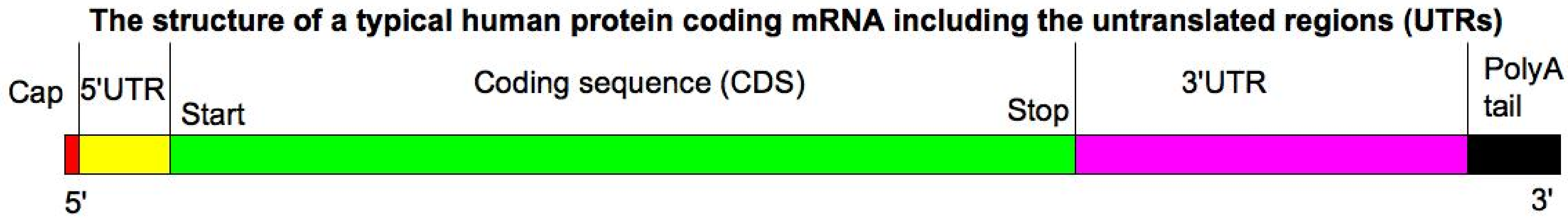
Molecular mechanisms of transcription activation and repression

1. Chromatin mediated transcription control
2. Transcription control through the Mediator
3. Epigenetic control through DNA methylation
4. Regulation of transcription factor activity
5. Nuclear receptor

Post-transcriptional events

（转录后加工）

Post-transcriptional events



1. mRNA Processing I:
   - Splicing (剪接)
2. mRNA Processing II:
   - Capping(加帽)
   - Polyadenylation (加尾)
3. Other RNA Processing: rRNA and tRNA

mRNA Processing I: Splicing

Genes in pieces (The Nobel Prize in Physiology or Medicine 1993)

- **Intron(内含子):** a region that interrupts the transcribed part of a gene.
- **Exon(外显子):** a region of a gene that is ultimately presented in that gene's mature transcript.

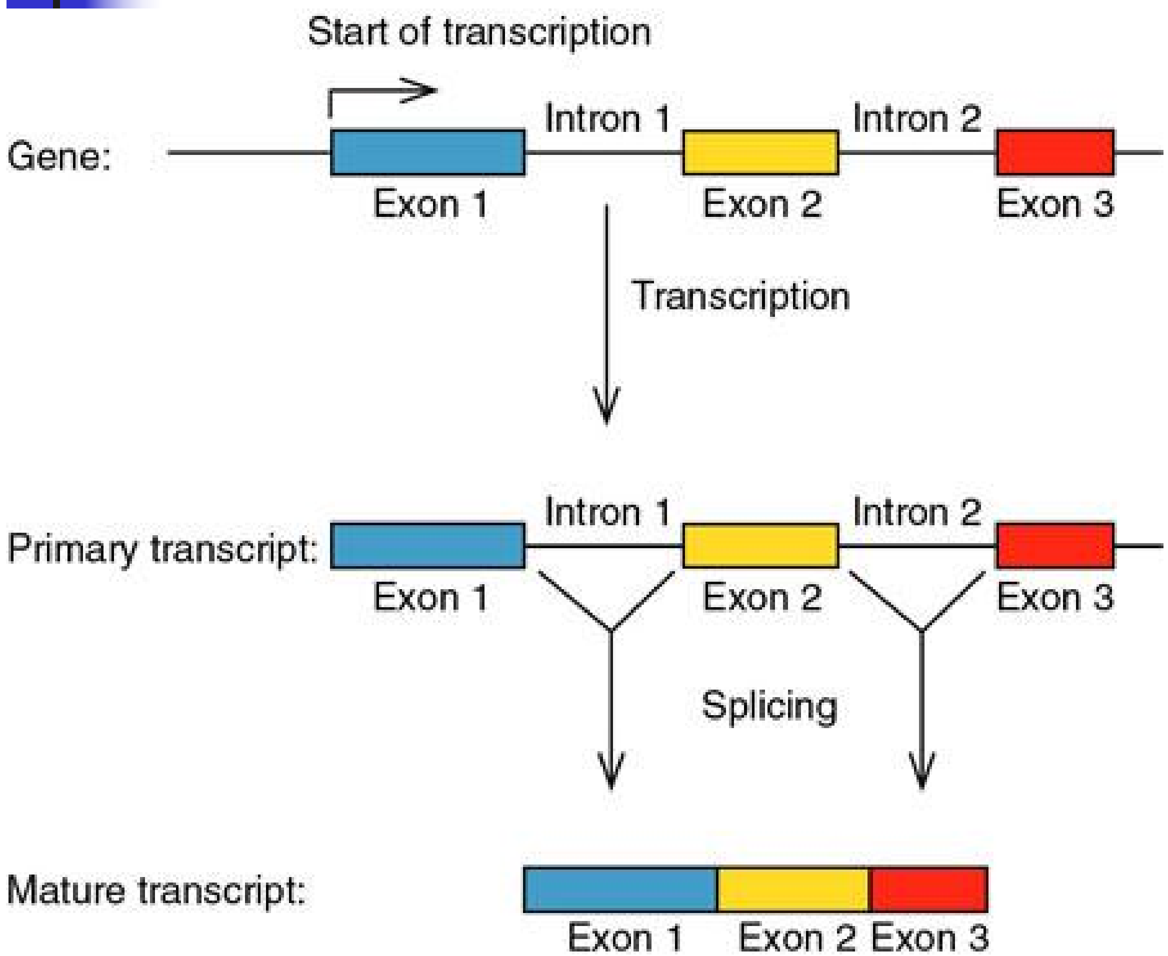
- **RNA splicing (剪接):** the process of cutting introns out of immature RNAs and stitching the exons together to form the final product

Most higher eukaryotic genes, coding for mRNA and tRNA, and a few coding for rRNA, are interrupted by unrelated regoins called introns.

The other parts of the gene, surrounding the introns, are called exons; The exons contain the sequences that finally appear in the mature RNA product.

Genes for mRNAs have been found with anywhere from zero to 60 introns. Transfer RNA genes have either zero or one.

mRNA Splicing (mRNA 剪接）



Introns are transcribed, then removed

Matured RNA

The Nobel Prize in Physiology or Medicine 1993 "for their discoveries of split genes"

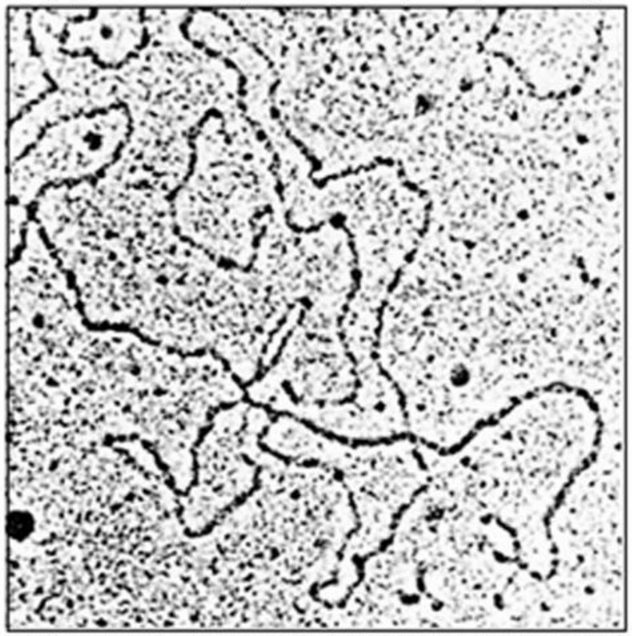
Richard J. Roberts

United Kingdom
New England Biolabs
Beverly, MA, USA
1943

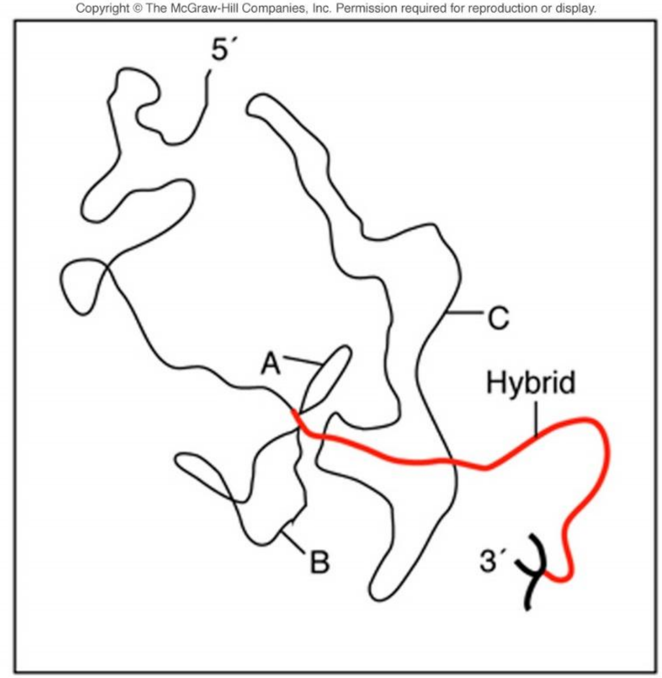
Phillip A. Sharp

Massachusetts Institute of Technology (MIT),
Center for Cancer Research
Cambridge, MA, USA
1944

Evidence for split genes

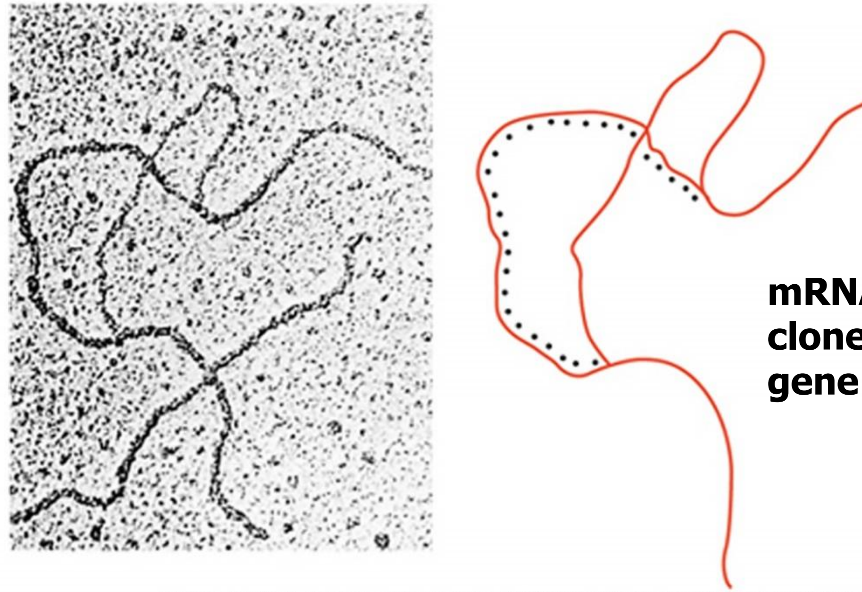


(Source: Berget, M., Moore, and Sharp, Spliced segments at the 3c terminus of adenovirus 2 late mRNA. *Proceedings of the National Academy of Sciences USA* 74: 3173, 1977.)



R-looping experiments reveal introns in adenovirus: a messenger RNA is hybridized to the DNA from which it had been transcribed. (*PNAS,1977)*

Hybridization between mRNA precursor and DNA. smooth , uninterrupted R-loop

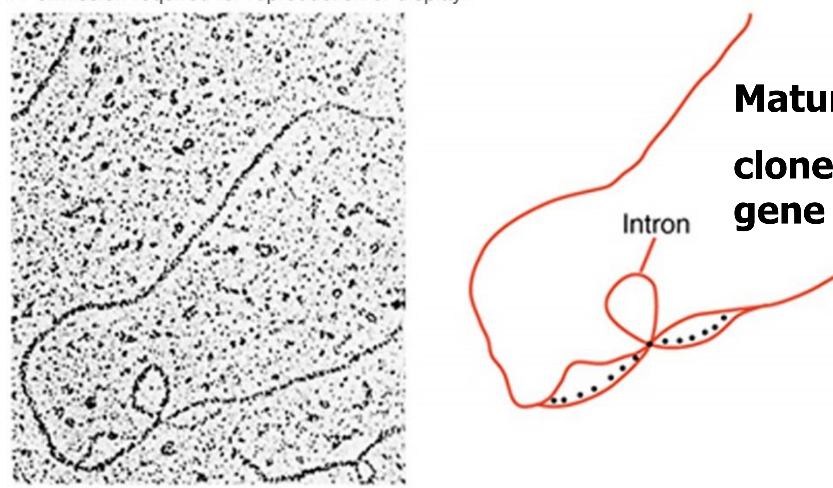


mRNA precursor+ cloned mouse ß –globin gene

(Source: Tilghman, S., Curtis, D., Tiemeier, P., and C. Weissmann, The intervening sequence of a mouse β-globin

Hybridization between mature mRNA and DNA

The large intron looped out, interrupted



Mature mRNA+

cloned mouse ß –globin gene

; transcribed within the 15S β-globin mRNA precursor. *Proceedings of the National Academy of Sciences USA* 75:1312, 1978.)

Summary

mRNA synthesis in eukaryotes occurs in stages. The 1st stage is synthesis of the primary transcription product, an mRNA precursor that still contains introns copied from the gene.

This precursor is part of a pool of large nuclear RNAs called hnRNAs. The 2nd stage is mRNA maturation. Part of the maturation is the removal of its introns in a process called splicing.

Splicing Signals

- First two bases of the intron: GU
- Last two bases of the intron: AG
- Following consensus sequence:

AG|GUAAGU – intron – YNCURAC – YnNAG|G (mammalian)

Y: pyrimidine, U or C;

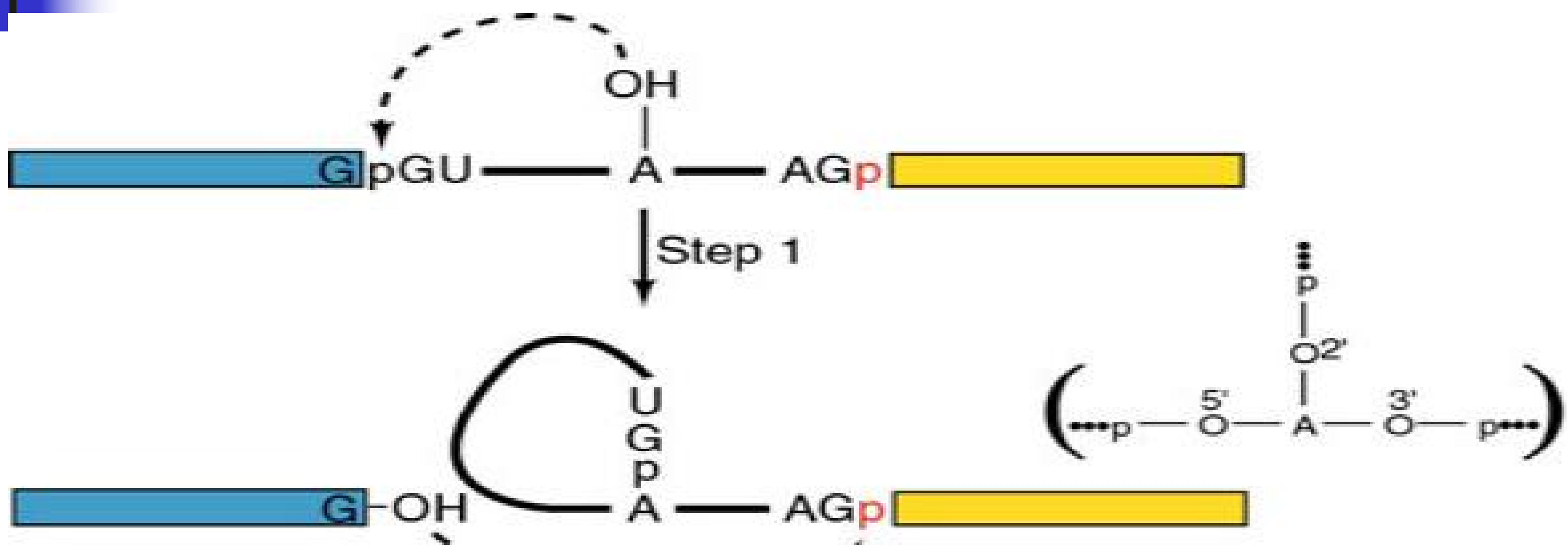
R: purine, A or G;

N: any base;

Yn: region of about nine pyrimidines

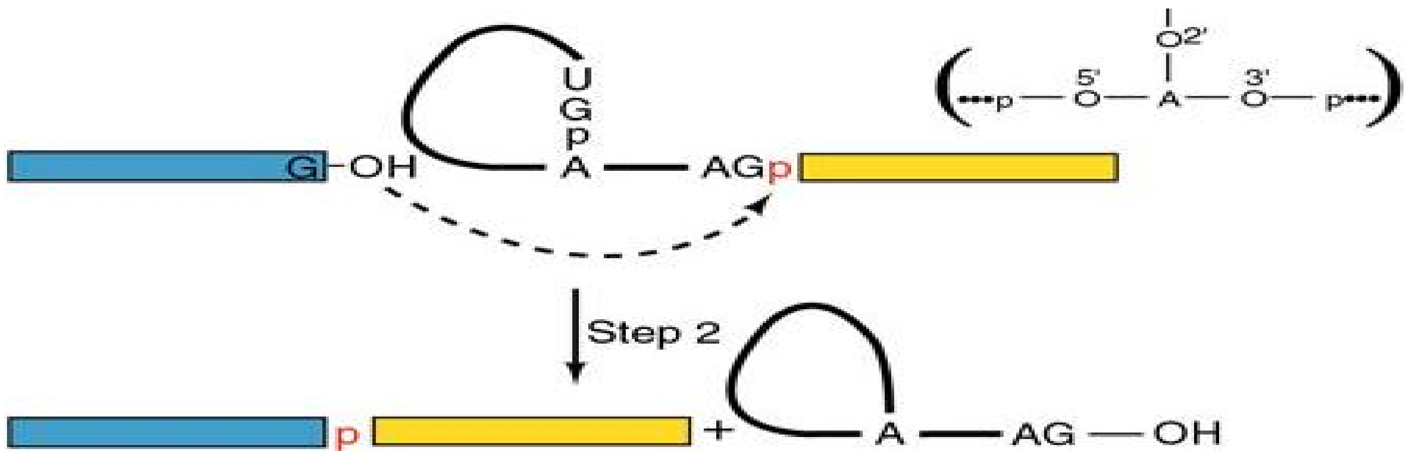
A: the branch point

Mechanism of Splicing

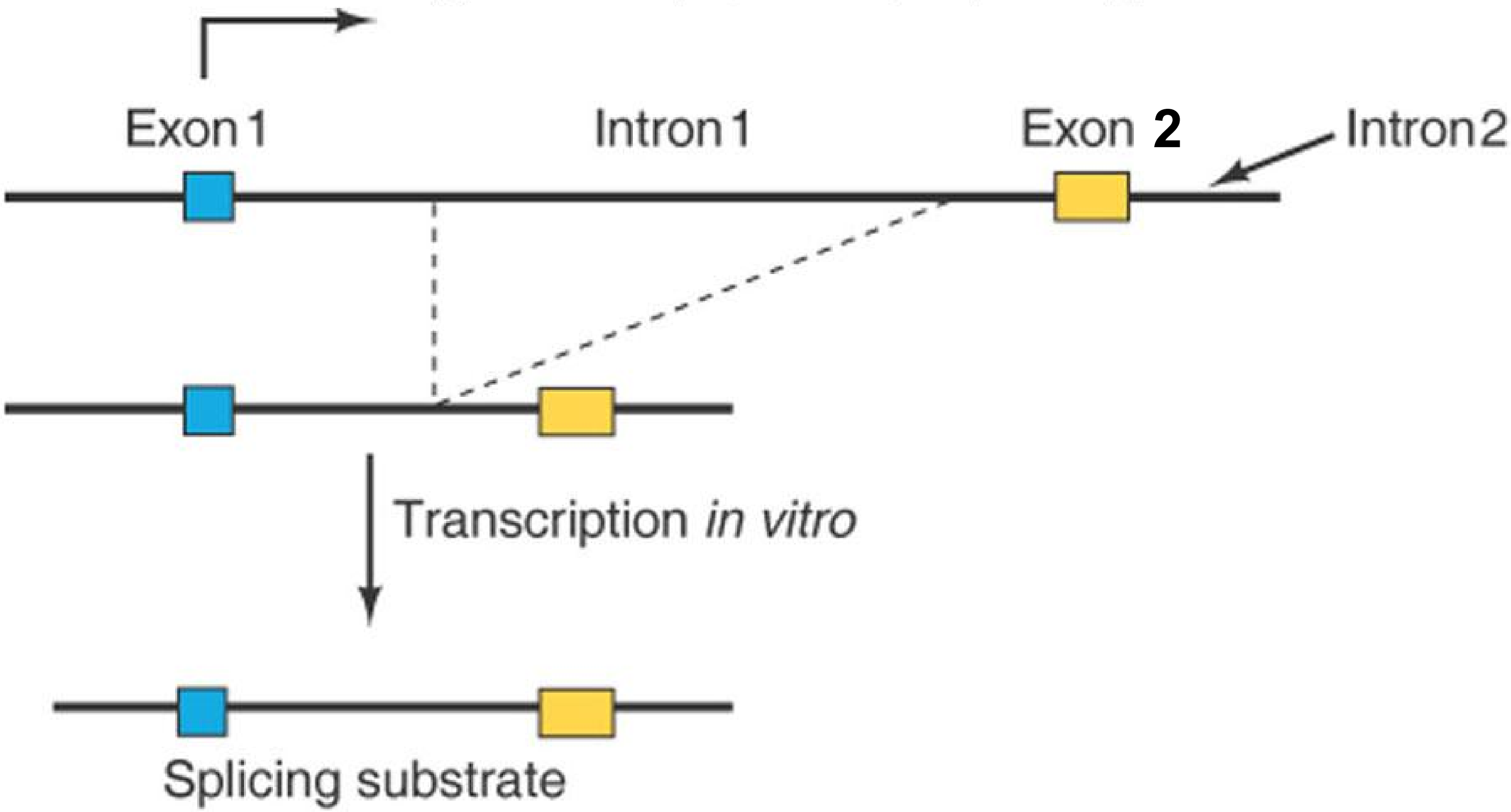


Step 1: **the 2’-hydroxyl group of the branch point A attacks on the phosphodiester bond of the 5’ exon/intron boundary, displacing the 5‘ exon (in blue) as the leaving group and giving the lariat (套索)structure shown in the middle.**

Mechanism of Splicing



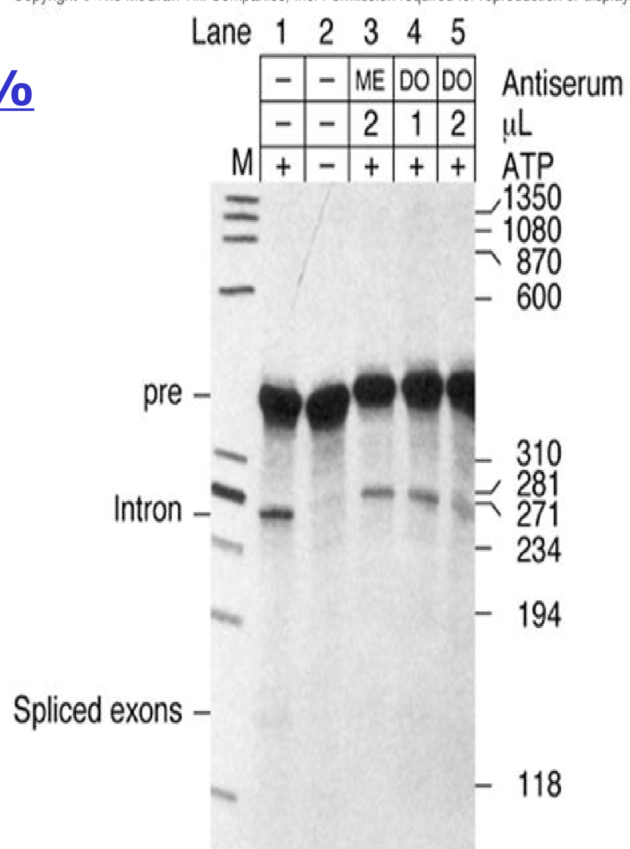
Step 2: **the 3' end of the 5' exon (in blue) attacks on the phosphodiester bond of the 3' intron/exon boundary, displacing the intron (as a lariat form) and sealing the two exons together (the blue and yellow).**



A splicing substrate Sharp and colleagues used.

Branched RNA (intron) move faster in 4% polyacrylamide gel and slower in 10% gel compared to the linearized RNA

4%



(Source: Grabowski P., R.A. Padgett, and P.A. Sharp, Messenger RNA splicing in vitro: an excised intervening sequence and a potential intermediate. Cell 37 (June 1984) f. 2, p. 417. Reprinted by permission of Elsevier Science.)

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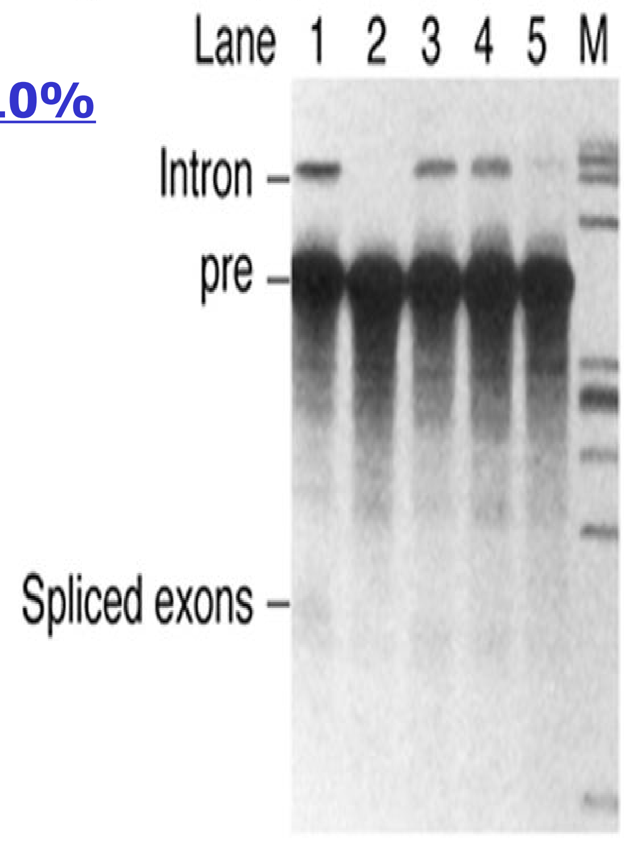
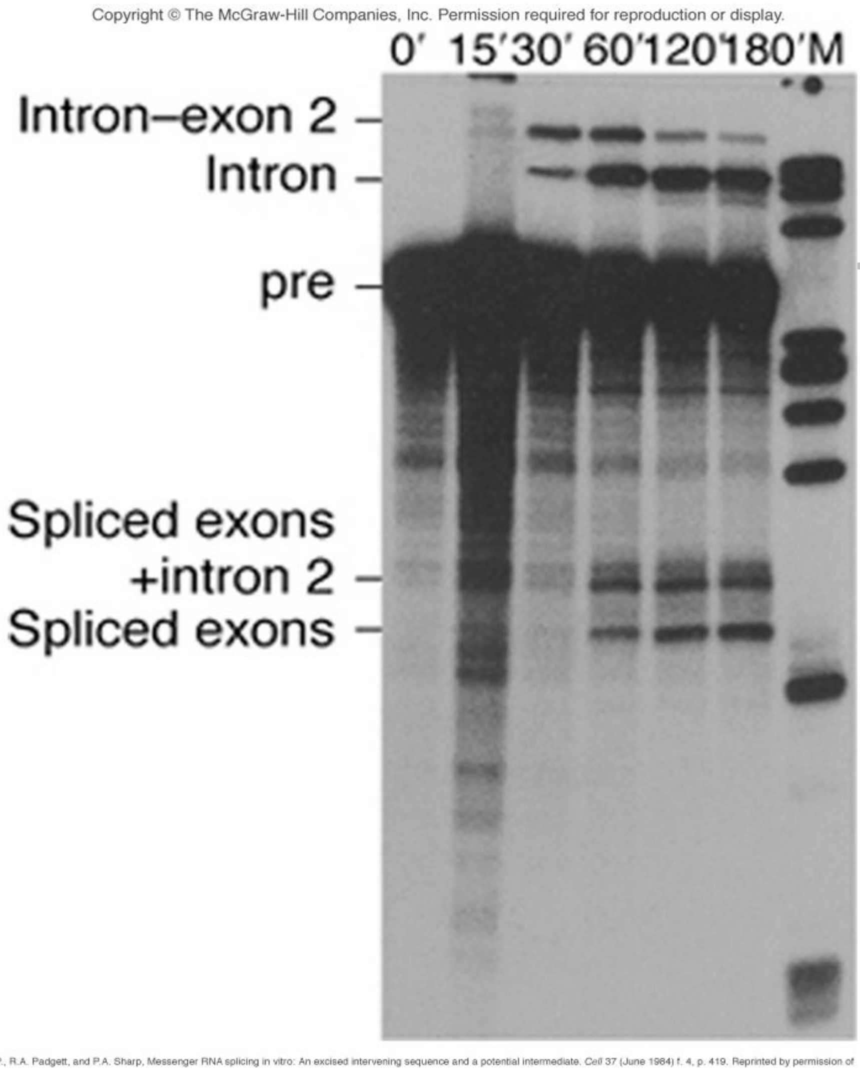
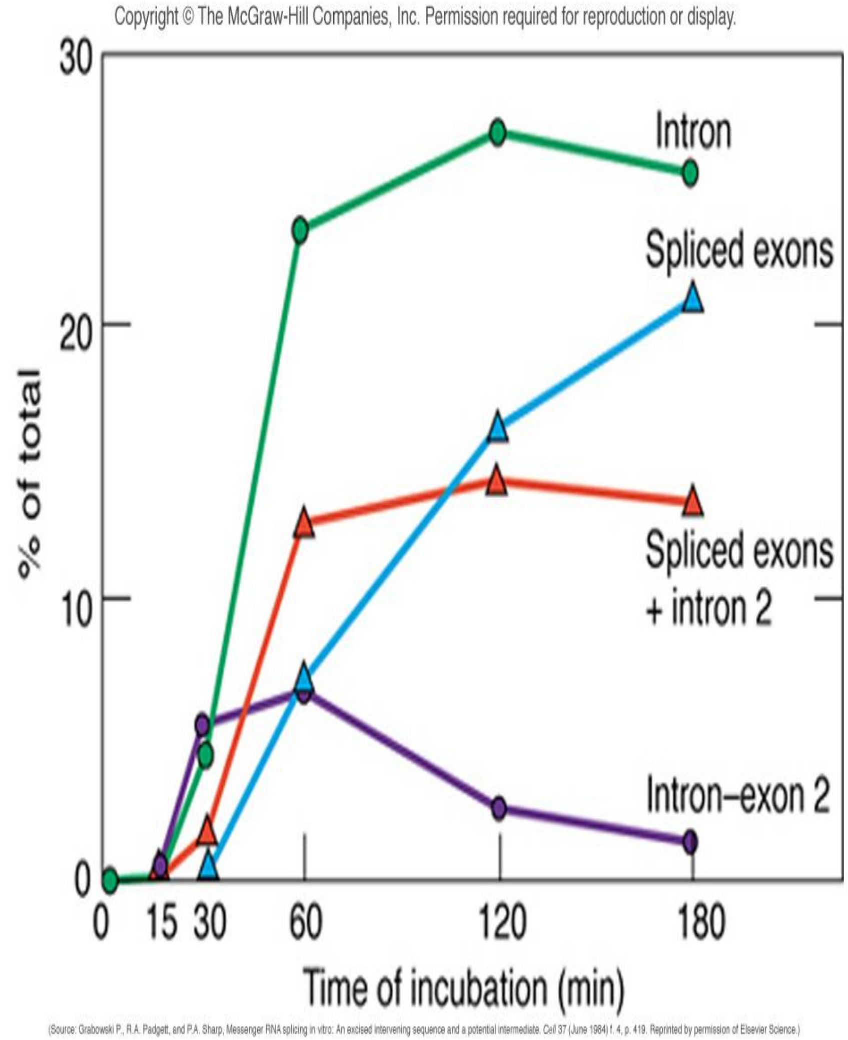
10%



P.A. Sharp, Messenger RNA splicing in vitro: an excised intervening sequence and a potential intermediate. Cell 37 (June 1984) f. 2, p. 417. Repri

Incubate the labeled substrate with Hela cell nuclear extract.





Time course of intermediate and liberated intron appearance (*Cell*, *1984* by Sharp)

Summary

Several lines of evidence demonstrate that nuclear mRNA precursors are spliced via a lariat-shaped, or branched, intermediate.

In addition to the consensus sequences at the 5'- and 3'- ends of nuclear introns, branch point consensus sequences also occur.

AG|GUAAGU – intron – YNCURAC – YnNAG|G

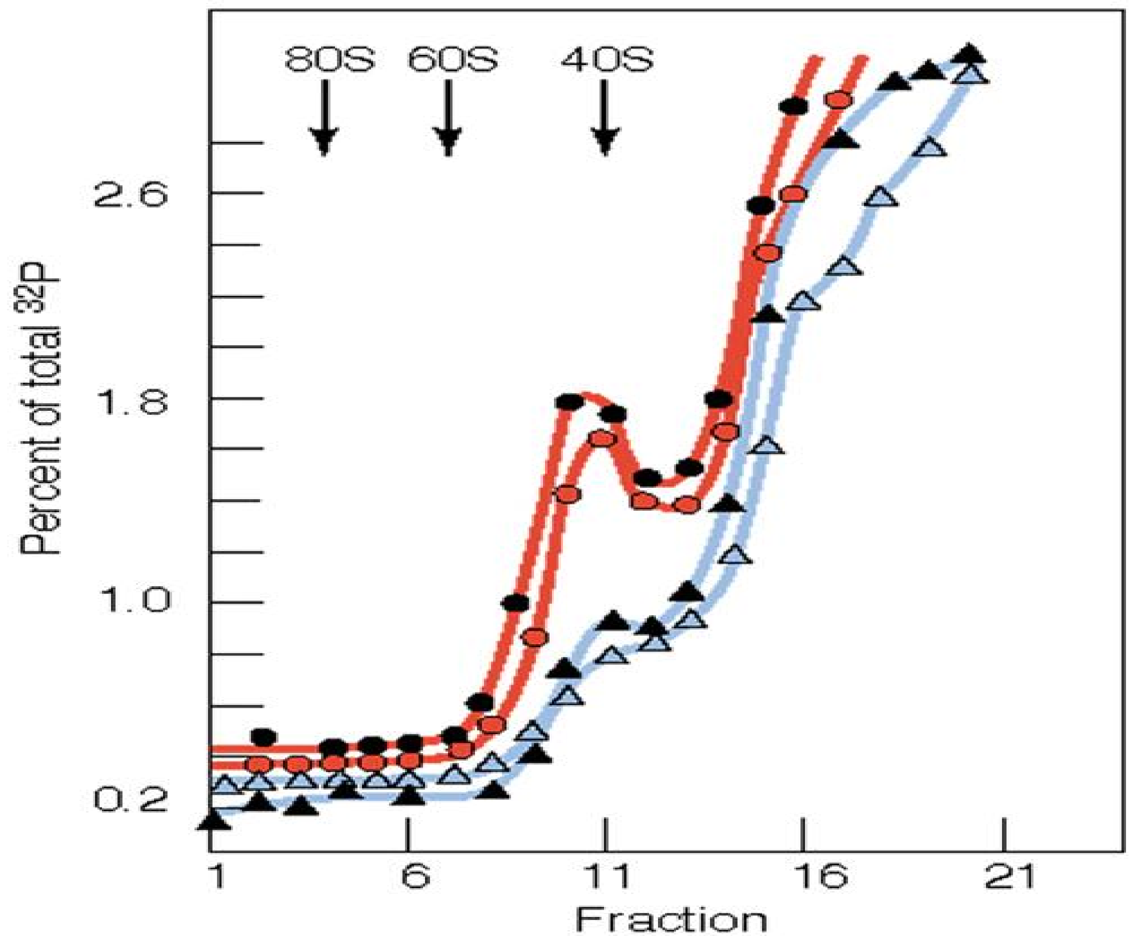
Spliceosome (剪接体)

Splicing takes place on a particle called spliceosome.

- **Spliceosome** : particle composed of RNA and peptide chains that affects mRNA splicing in the cell nucleus of eukaryotic cells
- **Structure of spliceosome:** SnRNA and proteins (SnRNPs, Snurps)
  - SnRNPs: small nuclear ribonucleoproteins；核小分子核糖核蛋白
  - Snurps: snRNAs +proteins

  Splicing involves variety of snRNP: U1, U2,U4,U5 and U6
- snRNAs can recognize critical splicing signals
- **U: uridine-rich RNA**

Yeast Spliceosome: 40S= RNA+protein



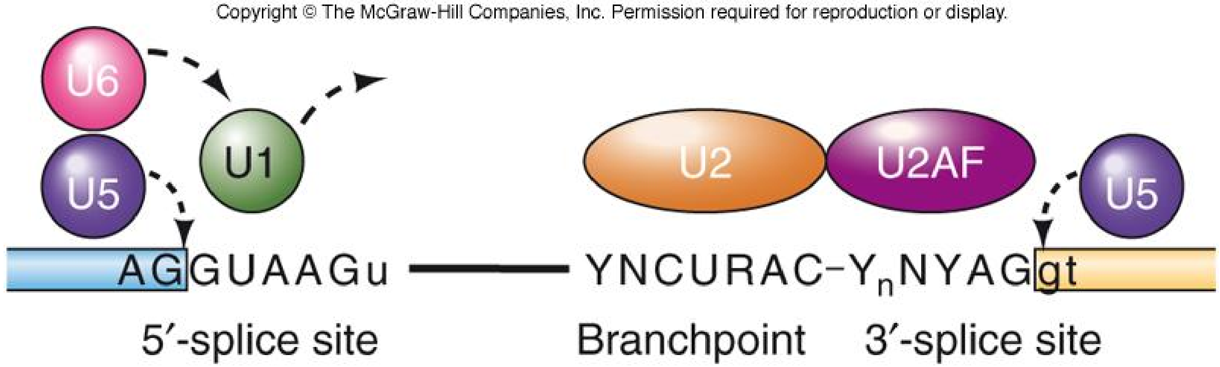
, The spliceosome: Yeast premessenger RNA associated with a 40S complex in a splicing-dependent reaction. Science 228:966, 1985. Copyright © 1985 American Association for the Advancemen

- Labeled pre-RNA was mixed with splicing extract, then glycerol gradient ultracentrifugation.
- Determine the radioactivity by scintillation.
- Red: WT
- Blue: Mutated at the 5'-splice site (A into C)``

(AAAS,1985, by Brody,E)

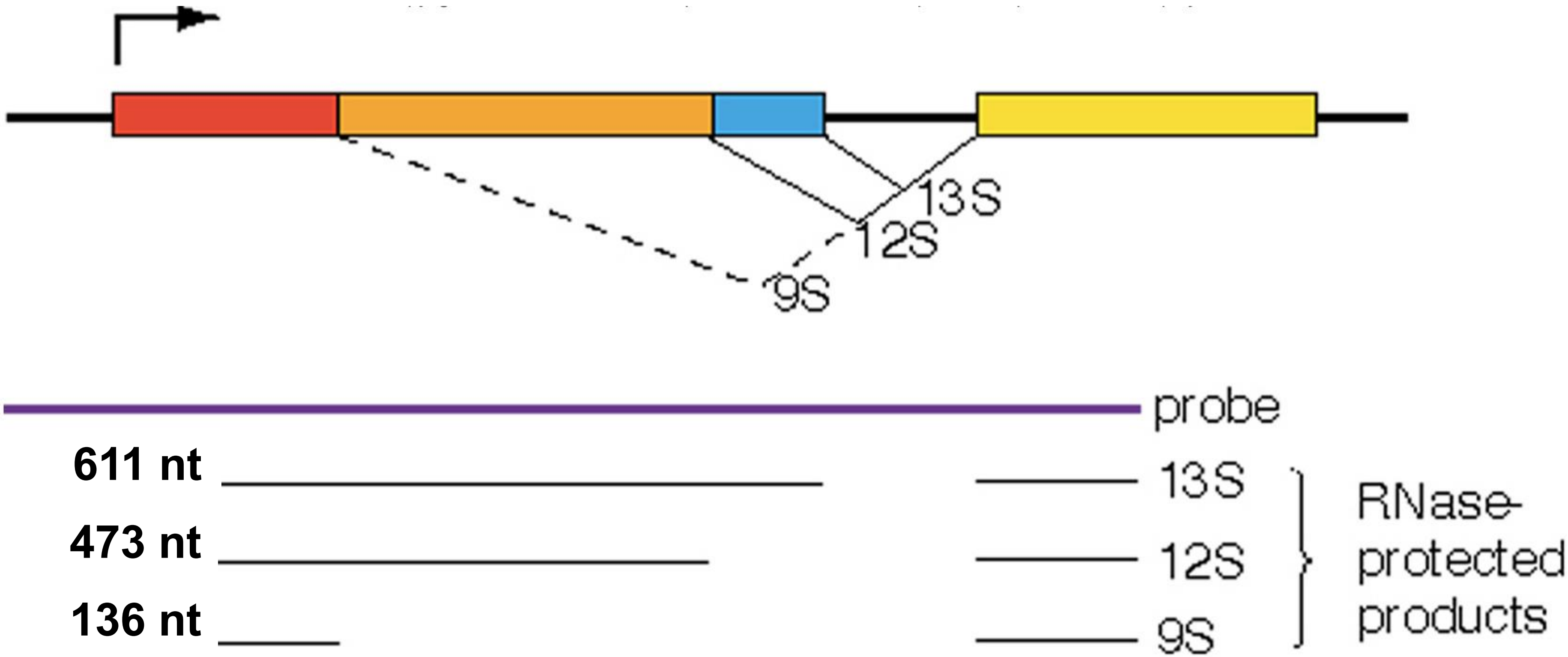
U1 snRNP

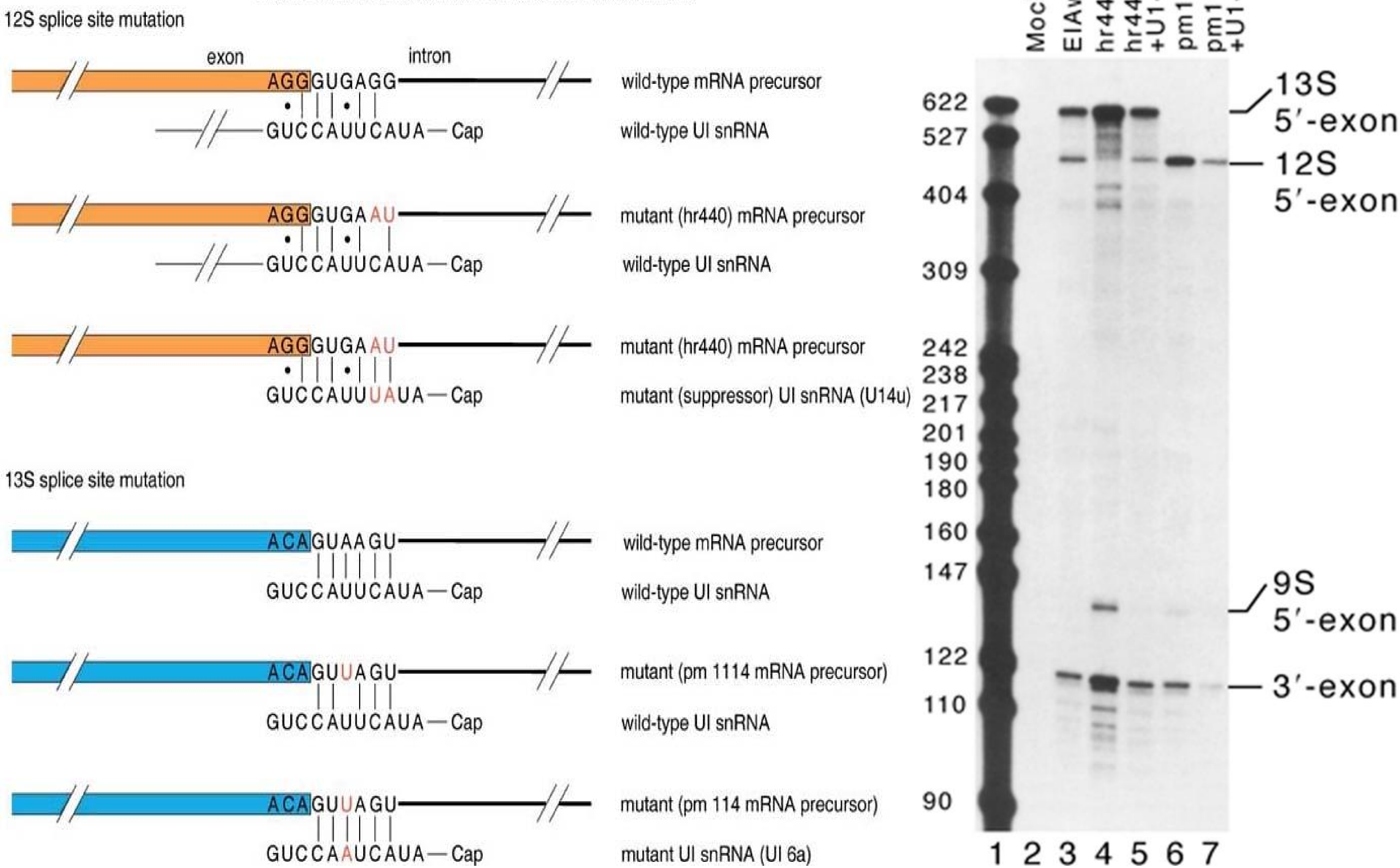
- U1 snRNA sequence is complementary to both 5'- and 3'-splice site consensus sequences
  - U1 snRNA base-pairs with these splice sites



Snurps: U1 snRNA

- Base pairing b/w U1 snRNA & the 5'-splice site of an mRNA precusor is necessary, but not sufficient for splicing

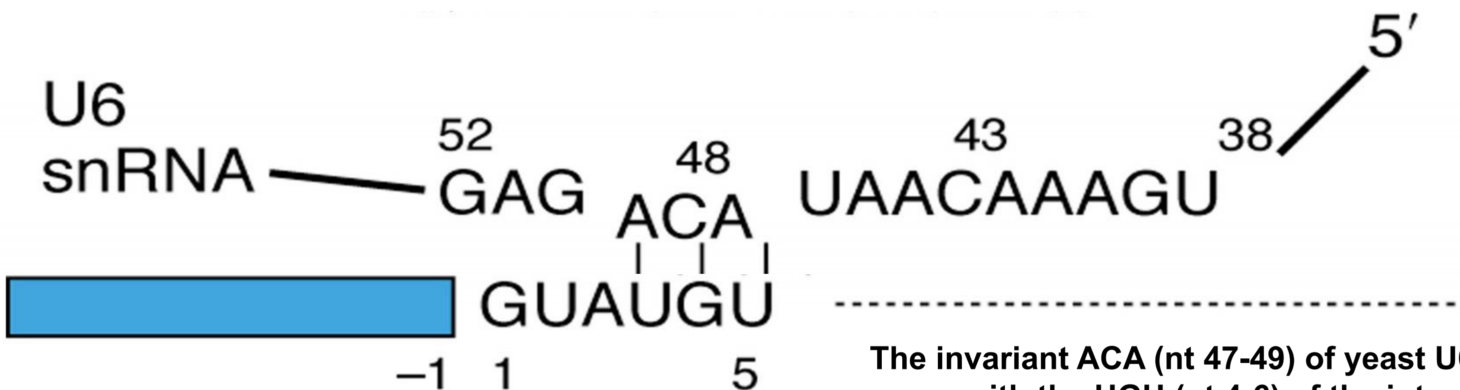




Alignment of wild-type and mutant 5'-splice sites with wild-type and mutant U1 snRNAs

U6 snRNP

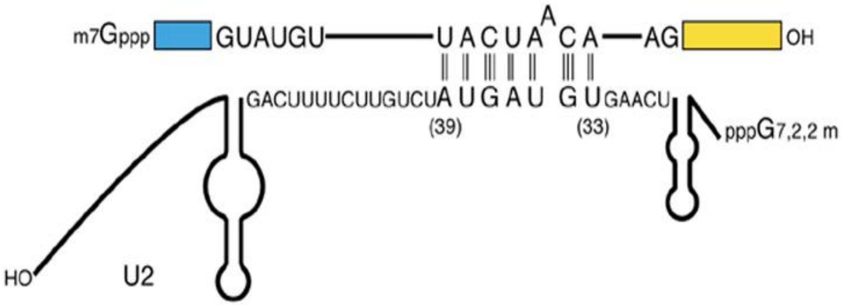
- It associates with the 5’-end of the intron by base pairing through U6 snRNA
- The base-pairing occurs prior to formation of the lariat intermediate. The association between U6 and the splicing substrate is essential for the splicing process.
- U6 also associates with U2 during splicing

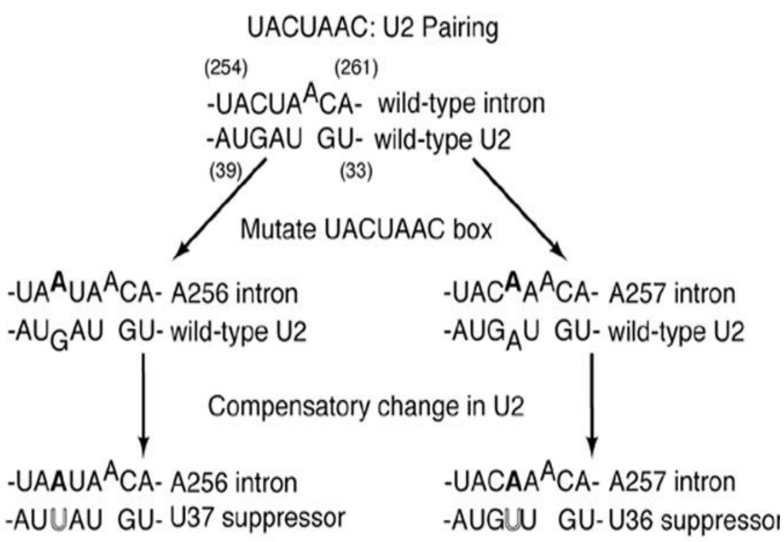


The invariant ACA (nt 47-49) of yeast U6 base-pairs with the UGU (nt 4-6) of the intron

U2 snRNP

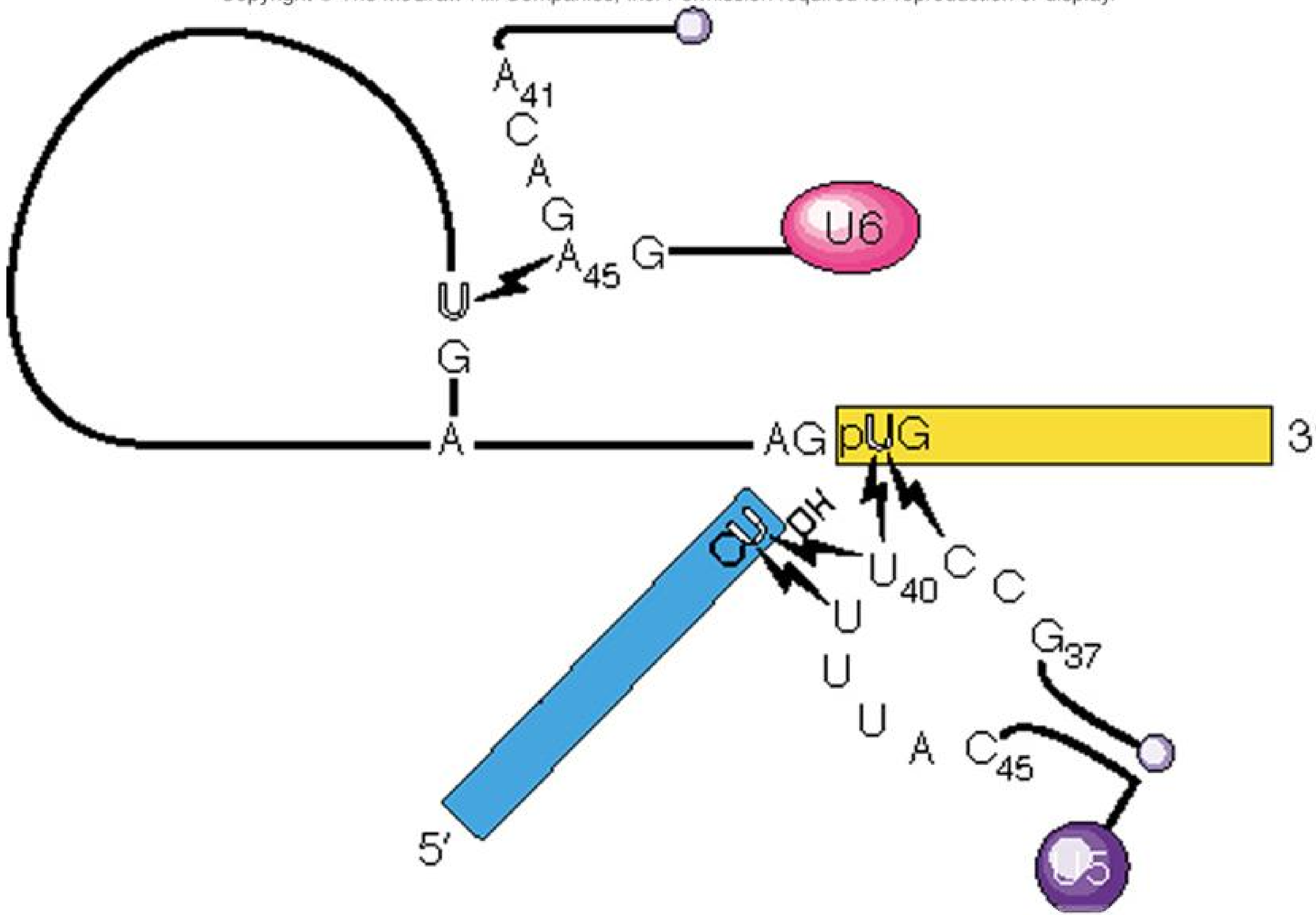
- U2 snRNA base-pairs with the conserved sequence at the splicing branchpoint.
- U2 also forms vital base pair with U6: forming a region called helix I (helps orient these snRNPs for splicing) & helix II(5'-end of U2 with 3'-end of U6, in mammalian)





U5 snRNP

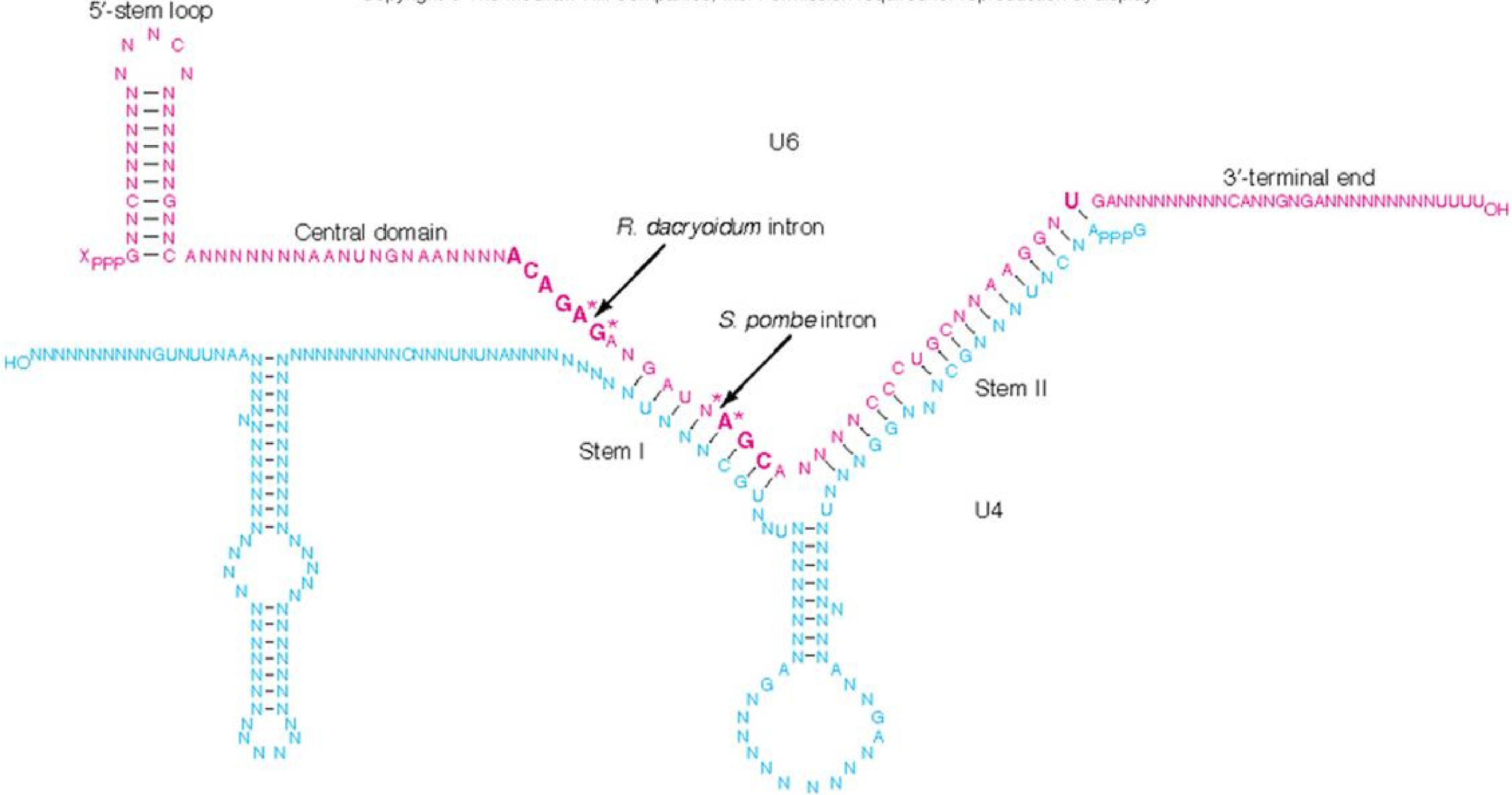
U5 snRNP associates with the last nucleotide in one exon and the first nucleotide of the next. This presumably lines the 2 exons up for splicing



(Source: From Sontheimer E.J. and J.A., Steitz, The U5 and U6 small nuclear RNAs as active site components of the spliceosome. *Science* 262:1995, 1993. Copyright © 1993 American Association for the Advancement of Science, Washington, DC. Reprinted by permission.)

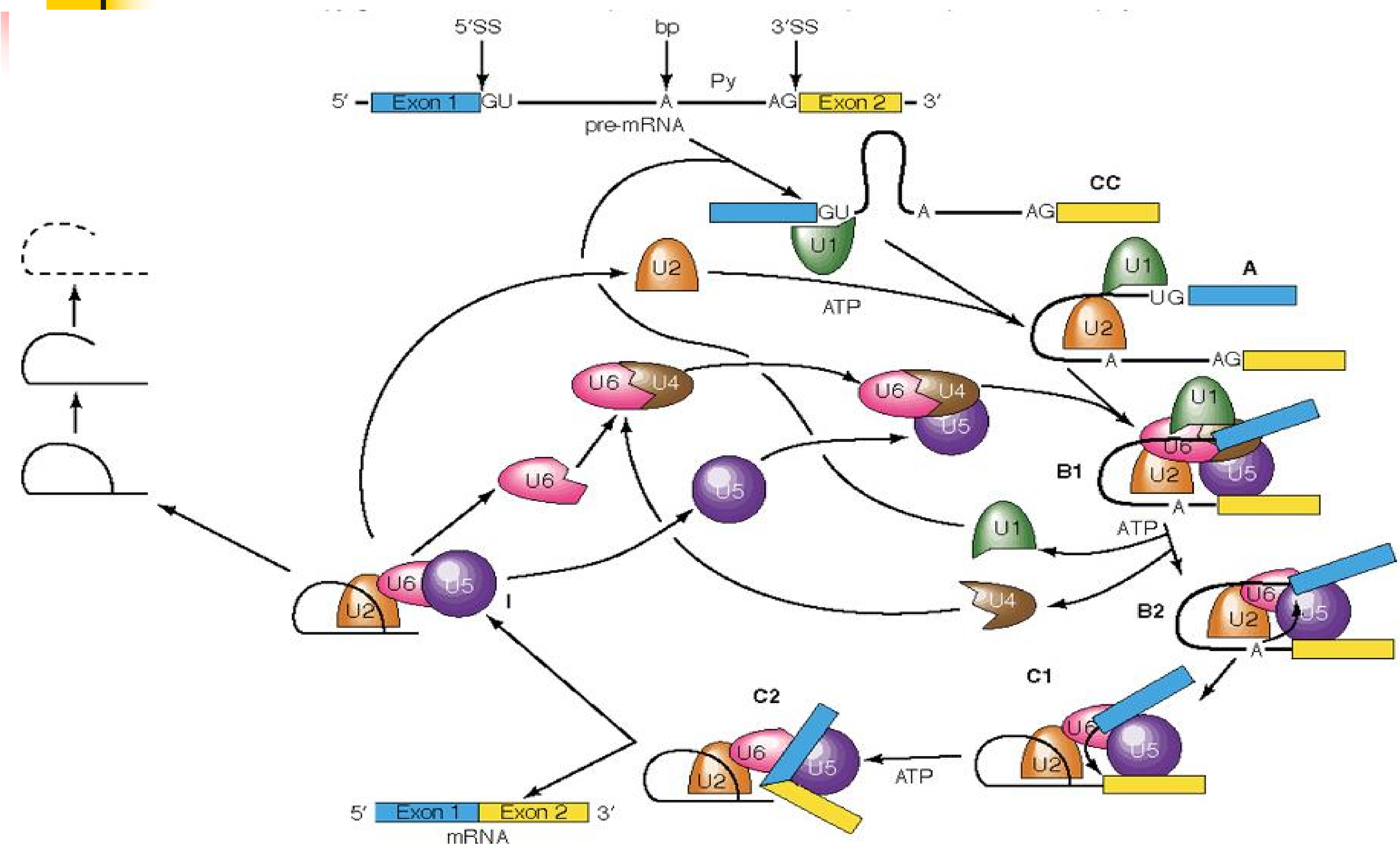
Summary of U5 and U6 interactions with the splicing substrates revealed by 4-thioU cross-linking

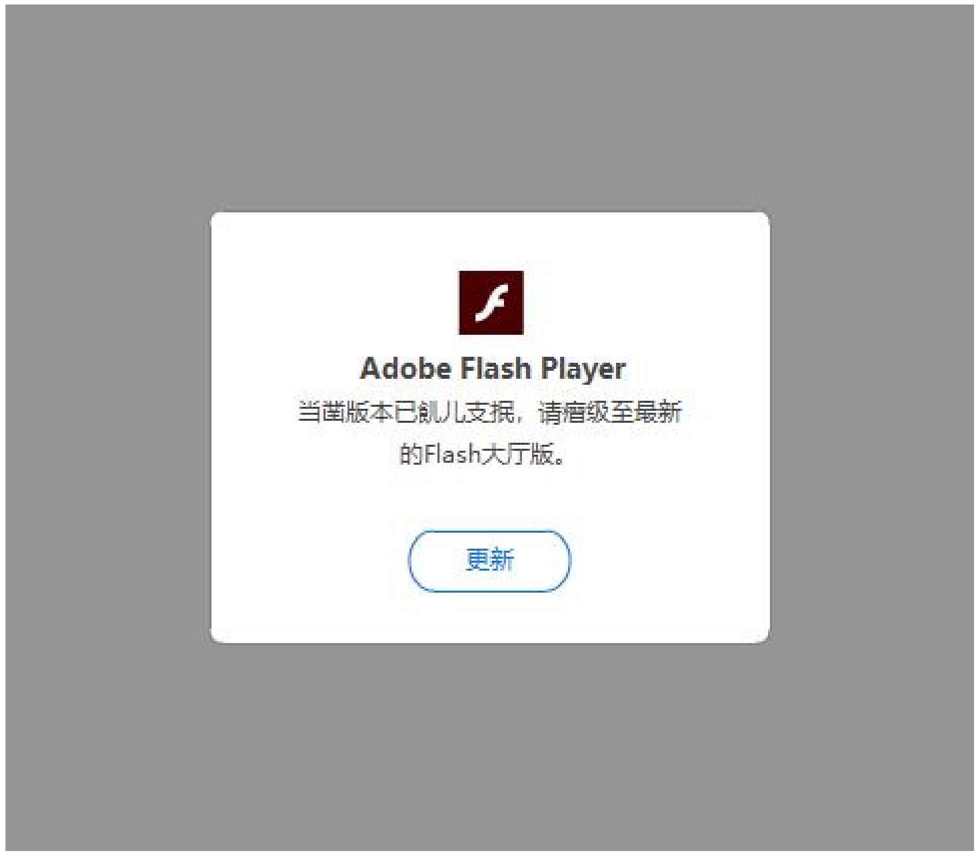
U4 snRNP: base-pairs with U6, it bind to U6 until U6 is needed in the splicing reaction



(Source: From Guthrie, C., Messenger RNA splicing in yeast: Clues to why the spliceosome is a ribonucleoprotein. *Science* 253:160, 1991. Copyright © 1991 American Association for the Advancement of Science, Washington, DC. Reprinted by permission.)

Model for the spliceosome cycle

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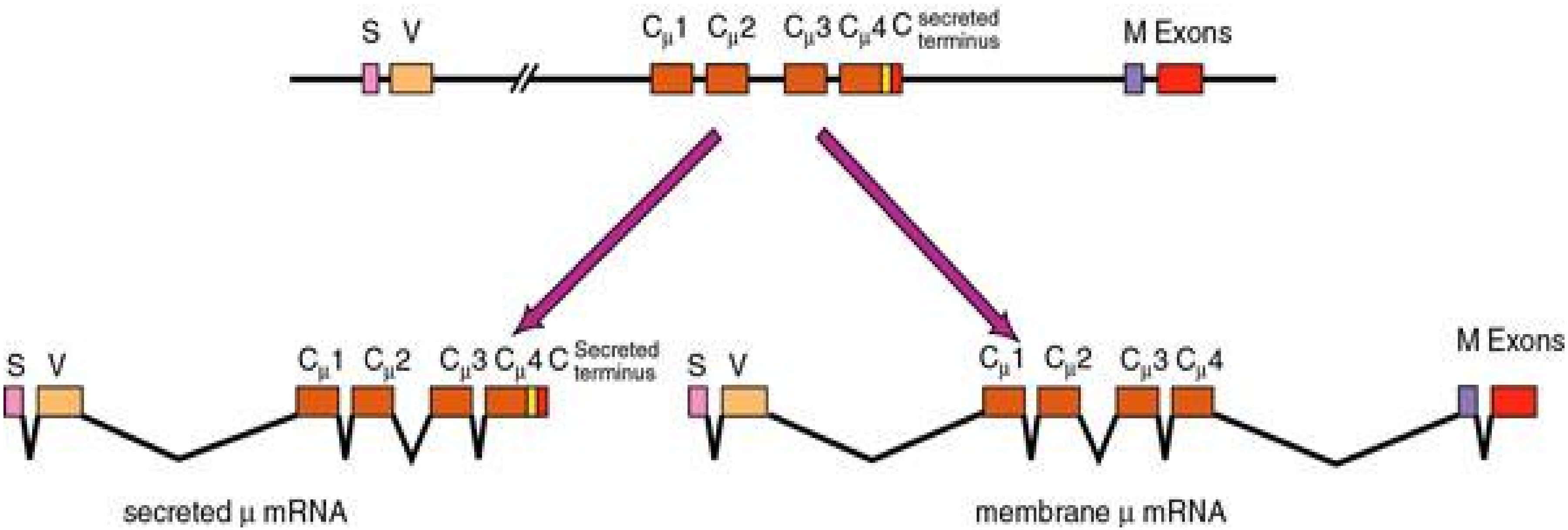
Alternative splicing

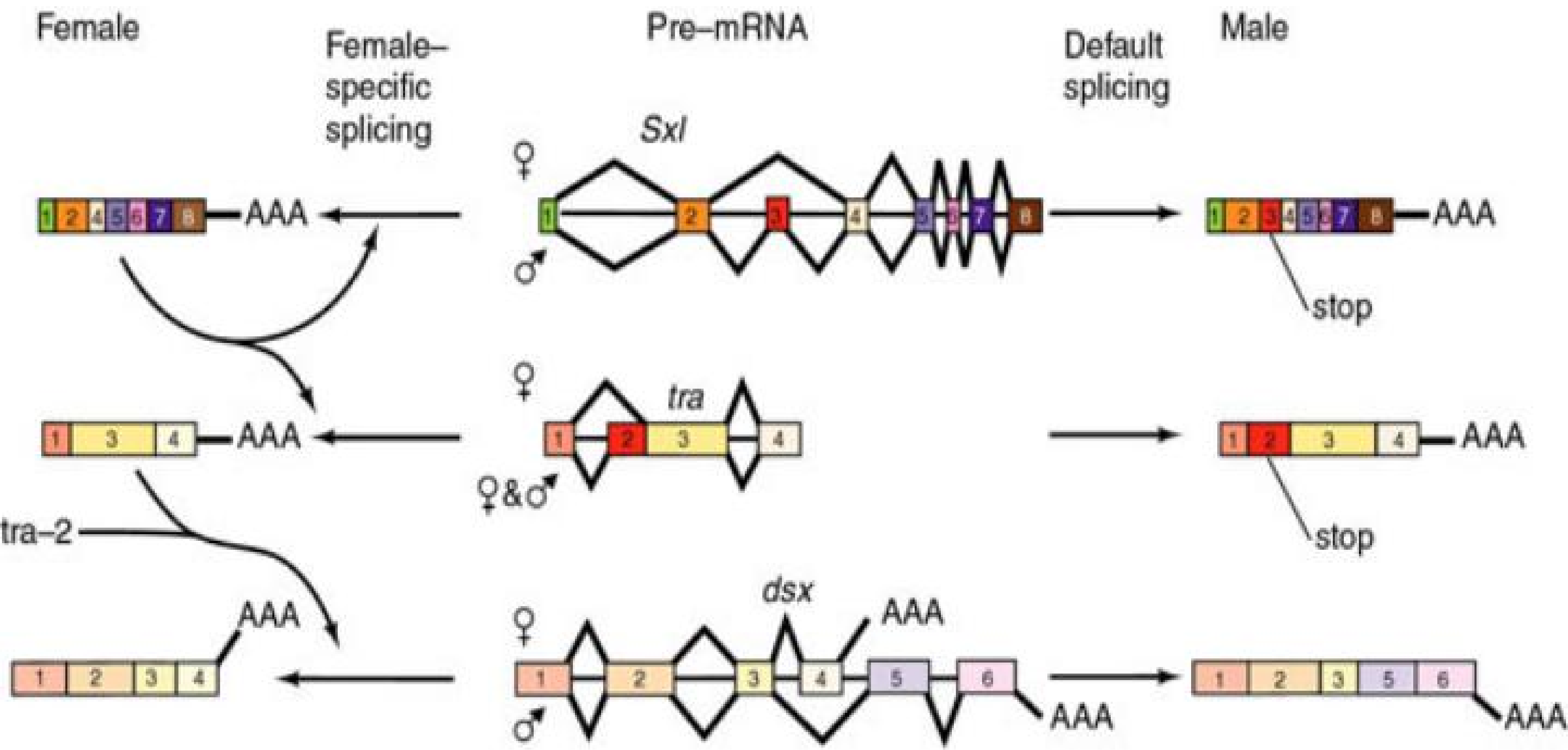
Alternative splicing: One precursor RNA can be spliced in different ways.

About 1 in 20 eukaryotic pre-mRNA can be spliced in more than one way. Leading to 2 or more alternative mRNAs that encode different proteins.

60% in Human

Examples: processing of a mRNA to make different parts of antibody molecules (immunoglobulins).





Sex determination pattern of Drosophila. There are three genes that are involved: *Sxl* (sex lethal), *tra* (transformer), and *dsx* (double sex). Each of these genes produces a pre-mRNA that has two possible splicing patterns, depending upon whether the fly is male (XY) or female (XX). Curved arrows indicate the positive effects on splicing

Summary

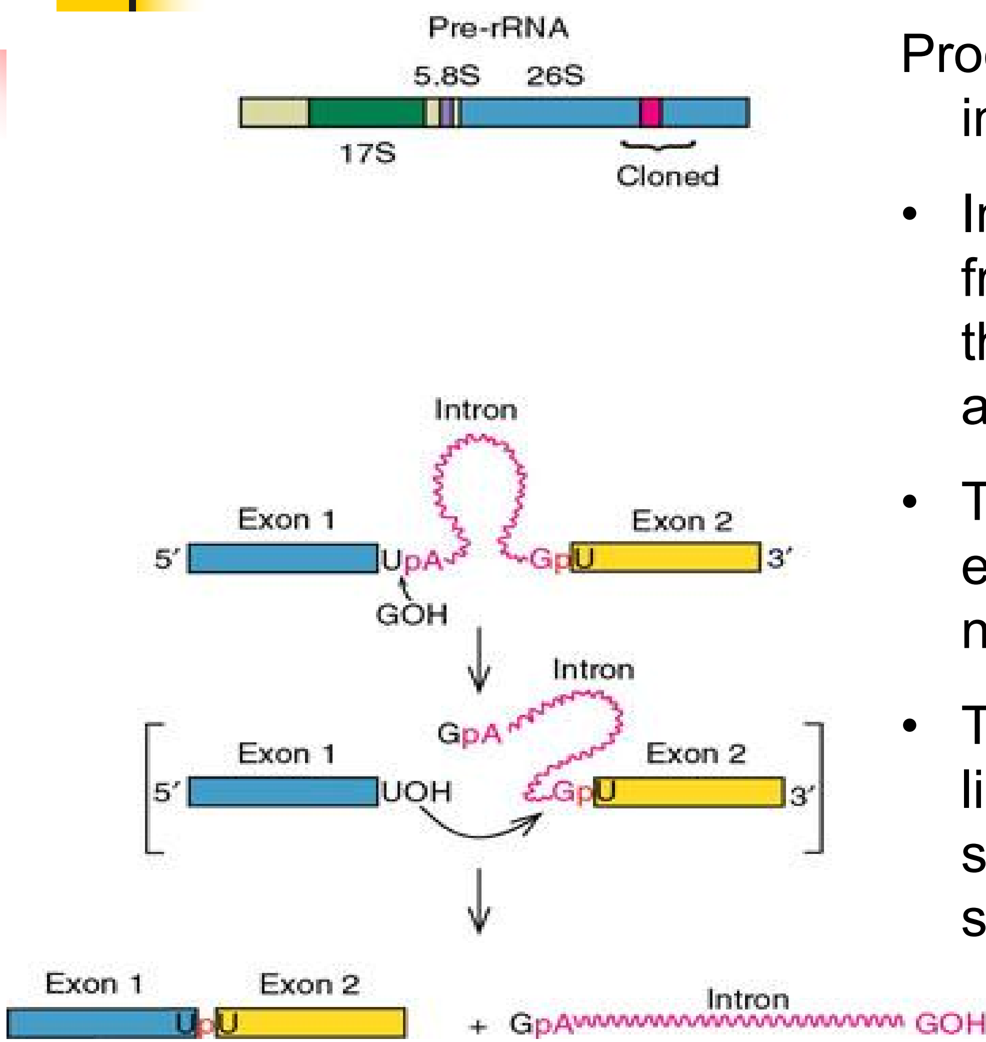
- The transcripts of many eukaryotic genes are subject to alternative splicing. This can have profound effects on the protein products of a gene.
- For example, it can make the difference between a secreted or a membrane-bound protein; it can even make the difference between activity and inactivity.

Self splicing RNAs

- Some RNAs could splice themselves w/o aid from spliceosome or any other proteins
- Two types of self splicing RNAs containing introns:
  - Group I introns   I型
  - Group II introns

The introns of fungal mitochondrial genes were originally classified as group I or group II according to certain conserved sequences they contained.

Group I introns



Processing of ribosomal RNA precursors in *Tetrahymena* (原生动物四膜虫).

- Instead of a branch point A, there is a free nucleotide (GMP, GDP or GTP) that acts as the initial attacking group at the 5' exon/intron boundary.
- The nucleotide does not require energy, but the 3'-OH is used for the nucleophilic attack.
- The excised intron comes out as a linear structure instead of the lariat seen in the other two (Group II self-splicing and nuclear pre-mRNA).

- Group I introns can be removed *in vitro* with no help from protein.
- The action begins with an attack by a guanine nucleotide on the 5’-splice site, adding the G to the 5’-end of the intron, and releasing the first exon.
- In the second step, the first exon attacks the 3’-splice site, ligating the two exons together, and releasing the linear intron.

Group II introns

The Group II introns are found in mitochondria and chloroplast genes.

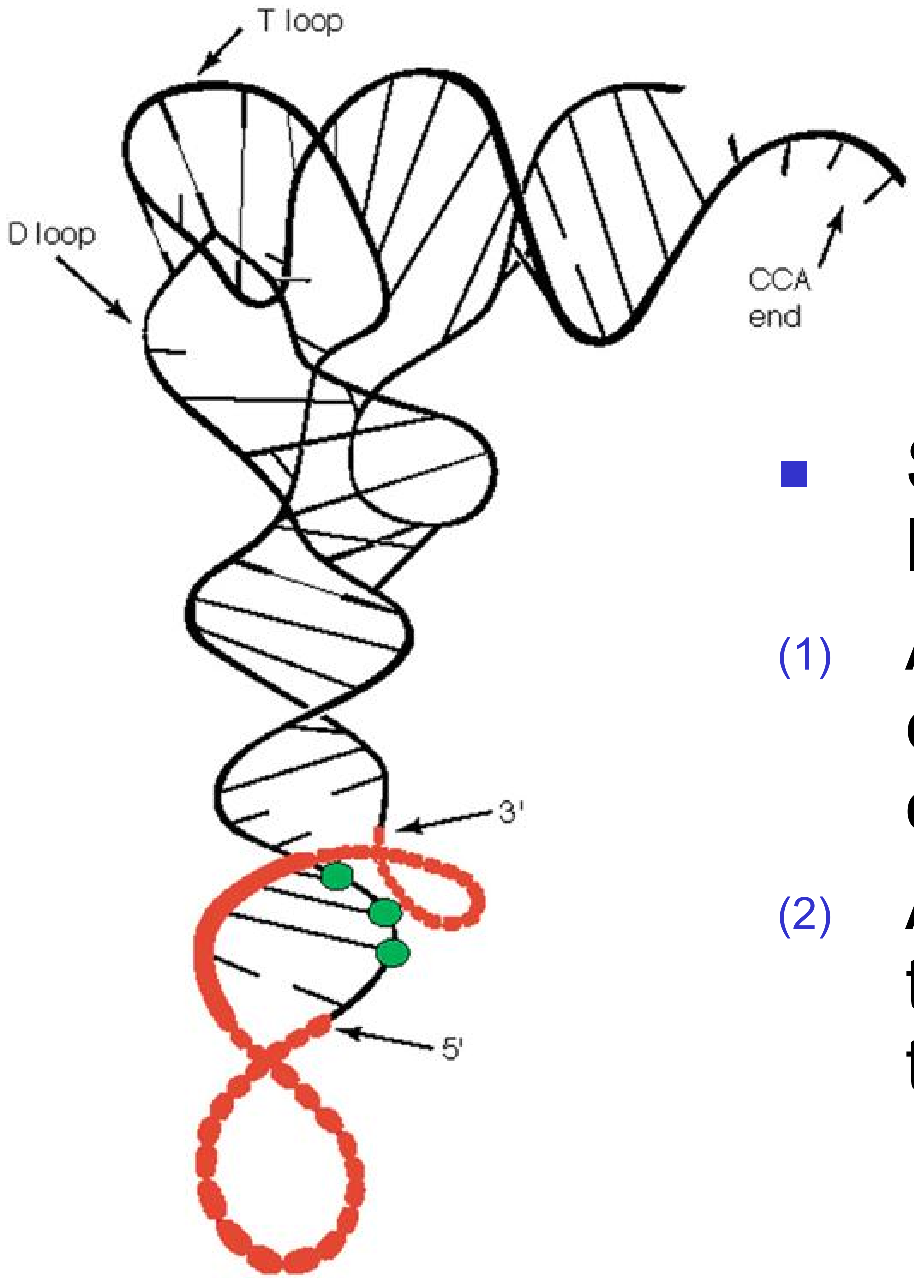
- The splicing mechanism used here is very much like that used in the nuclear pre-mRNA splicing we have been considering
- Except that there is no protein involved, not require energy input
- Basically, the number of phosphodiester bonds does not change. The only thing that changes is the position of the phosphodiester bonds

tRNA Splicing

- All the examples of splicing we have encountered so far have two steps:

(1) An attack on the 5'-splice site by a nucleotide in the intron or by a free nucleotide.

(2) An attack by the first exon on the second exon, which splices the two together.



- Splicing of tRNA also have two steps:

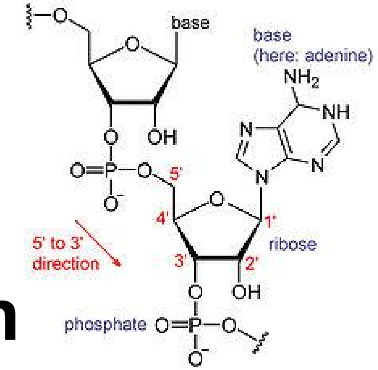
(1) A tRNA endonuclease cuts out the intron.

(2) A RNA ligase stitches the two half-tRNAs together.

(Source: From Lee M.-C. and G. Knapp, Transfer RNA splicing in *Saccharomyces cerevisiae*. *Journal of Biological Chemistry* 260:3113, 1985. Copyright © 1985 The American Society for Biochemistry & Molecular Biology, Bethesda, MD. Reprinted by permission.)

mRNA Processing II:

Capping & Polyadenylation

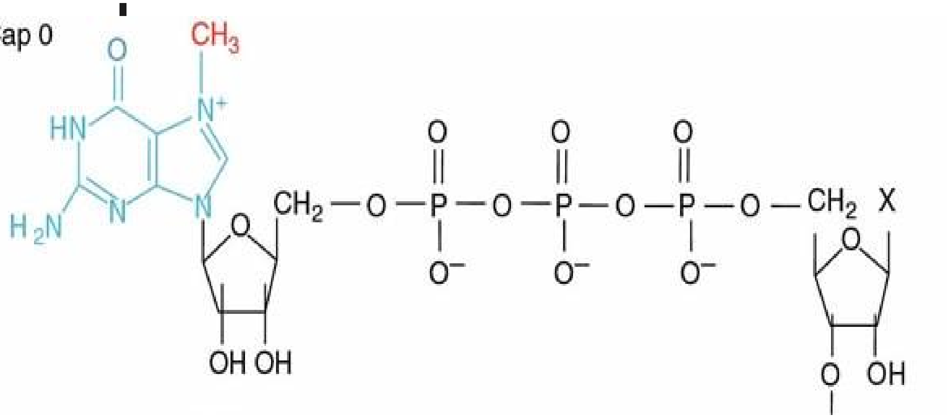


- **Capping:** A methylated guanosine bound through a 5’-5’ triphosphate linkage to the 5’-end of an mRNA. Happen in nucleus.

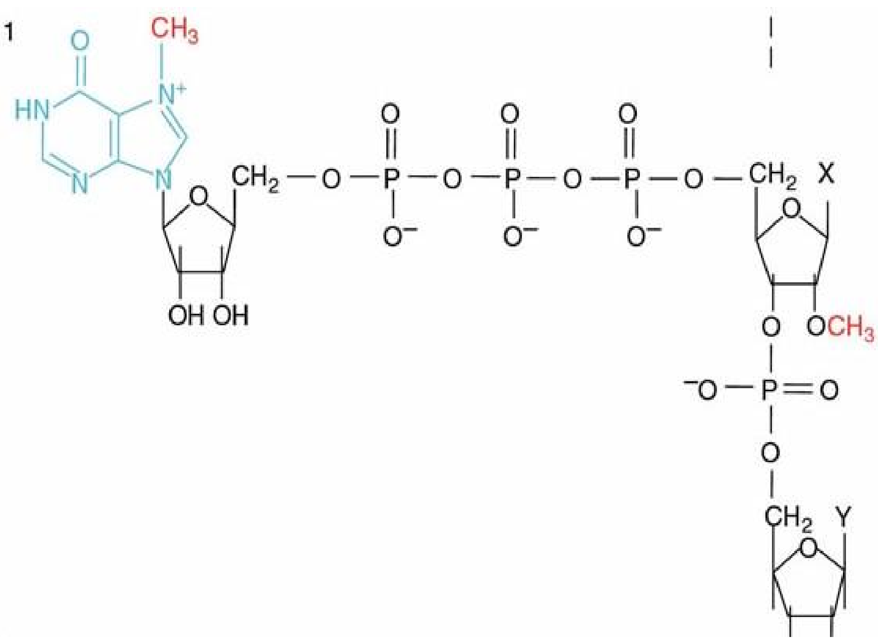
- **Types of capping:**
  1) Cap 0: 7-methyl-guanosine (m^7G), lack of 2'-O-methylation
  2) Cap 1: 7-methylated guanine & a 2'-O-methyl on this sugar of the first base
  3) Cap 2: 7-methylated guanine & 2'-O-methyls on the sugars of the first and second bases

Three cap structures

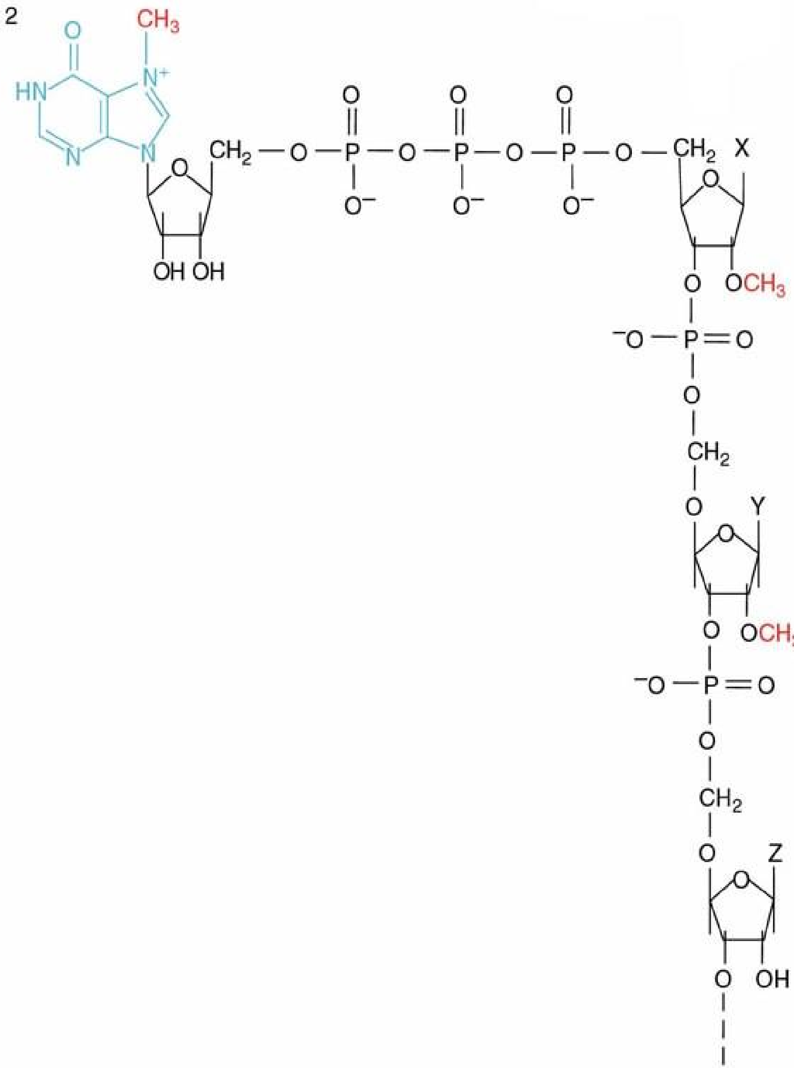
Cap 0



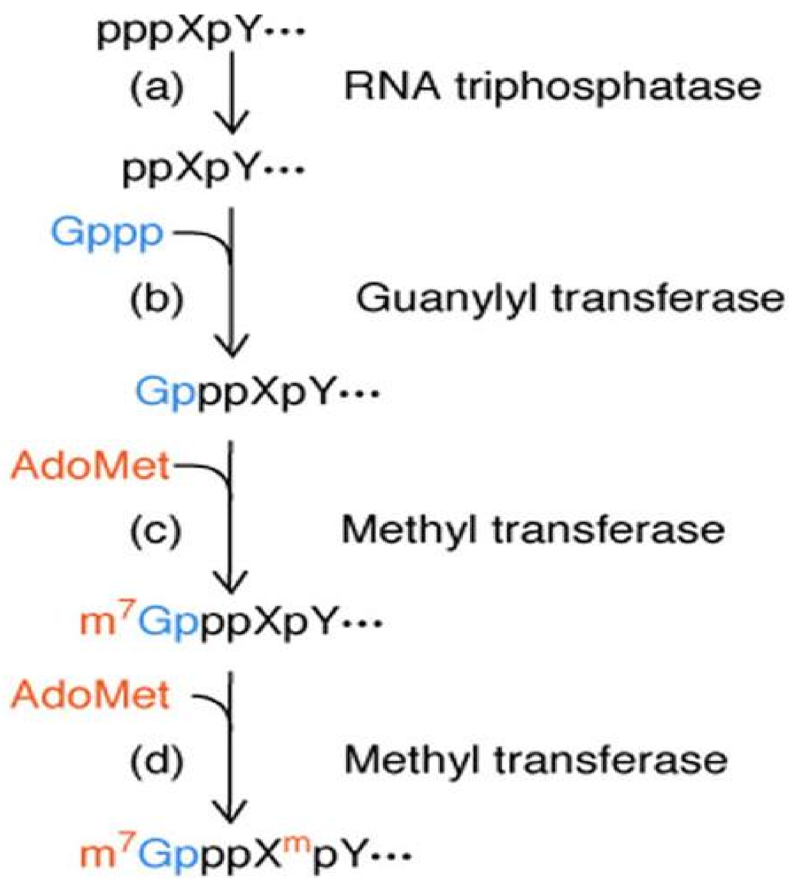
Cap 1



Cap 2



Cap synthesis



Sequence of events in capping

AdoMet: S-adenosylmethionine

1. γ-Phosphate of the base at the 5' end of the pre-mRNA is removed
2. Guanylyl transferease adds the GMP part of GTP to form a triphosphate linkage
3. The guanine residue of the cap is methylated at the N7 position of the base
4. Another methyl transferases add methyl groups to the 2'-OH of the ribose sugars of the next nucleotides

Cap’s Functions

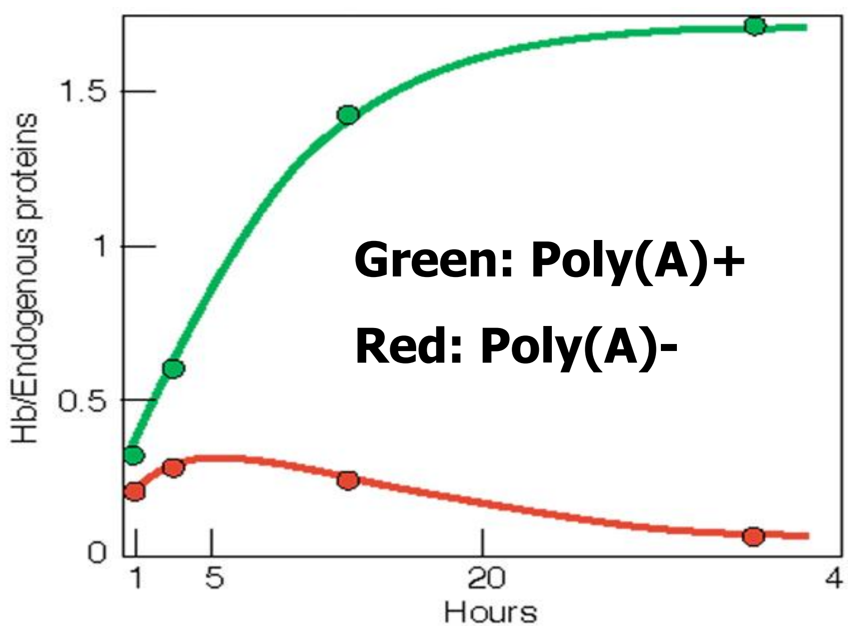
- Protection of the RNA from degradation
- Enhancement of the mRNA’s translatability. Increasing efficiency of translation
- Facility of mRNA transport from nucleus to cytoplasm
- Proper splicing of the pre-mRNA

3’ Polyadenylation

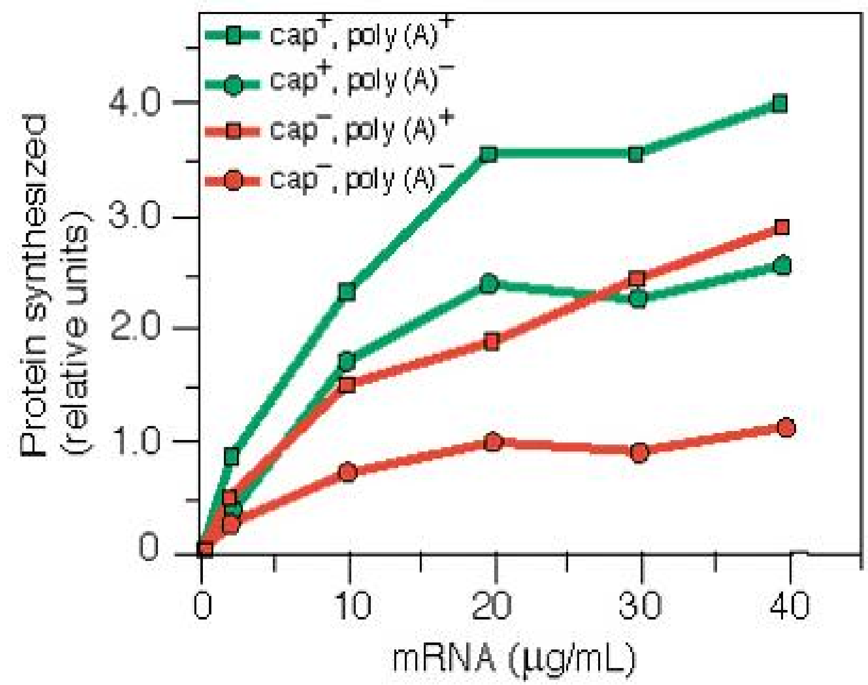
- Polyadenylation: A process of adding poly(A) to RNA
- Polyadenylation occurs post-transcriptionally by poly(A) polymerase
- About 250 nucleotides long
- Consensus sequence: AAUAAA

Function of poly (A)

- **Protection of mRNA**
- **Enhancement of the mRNA's translatability**



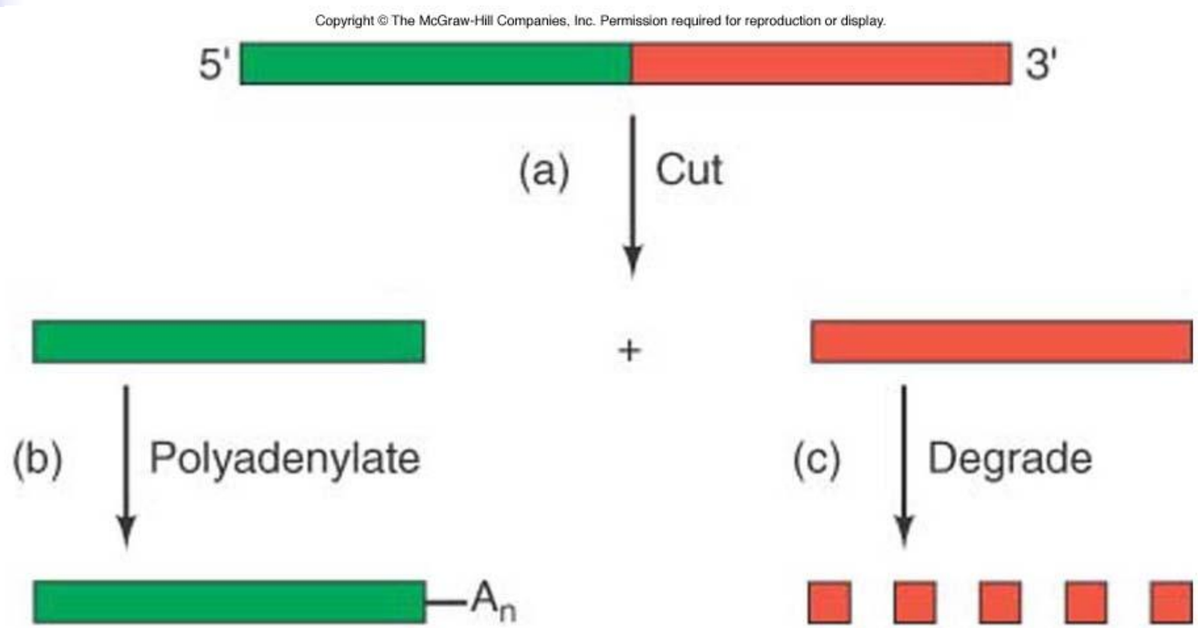
I. Leclercq, U. Nudel, H. Soreq, R. Solomon, B. Lebleu, M. Revel, and U.Z. Littauer, Role of the polyadenylate segment in the translation of globin messenger RNA in *Xenopus* oocytes. *Proceedings of the National Acad*



Effect of cap on RNA translatability (LUC)

	mRNA half life (min)	Enzyme activity	Effect of Poly (A)	Effect of cap
uncapped				
Poly(A)-	31	2941	1	1
Poly(A)+	44	4480	1.5	1
capped				
Poly(A)-	53	62595	1	21
Poly(A)+	100	1331917	21	297

Polyadenylation process



a, Cutting, cleaving the transcript. The cut occurs at the end of the RNA region
b, Polyadenylation. The poly(A) polymerase adds poly(A) to the 3’-end.
c, Degradation of the extra RNA.

Other RNA Processing: rRNA and tRNA

The trimming of excess regions from an RNA precursor is called processing.

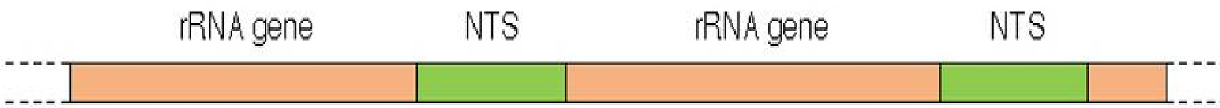
RNA processing is similar to splicing in that unnecessary RNA is removed, but differs from splicing in that no RNAs are stitched together during processing.

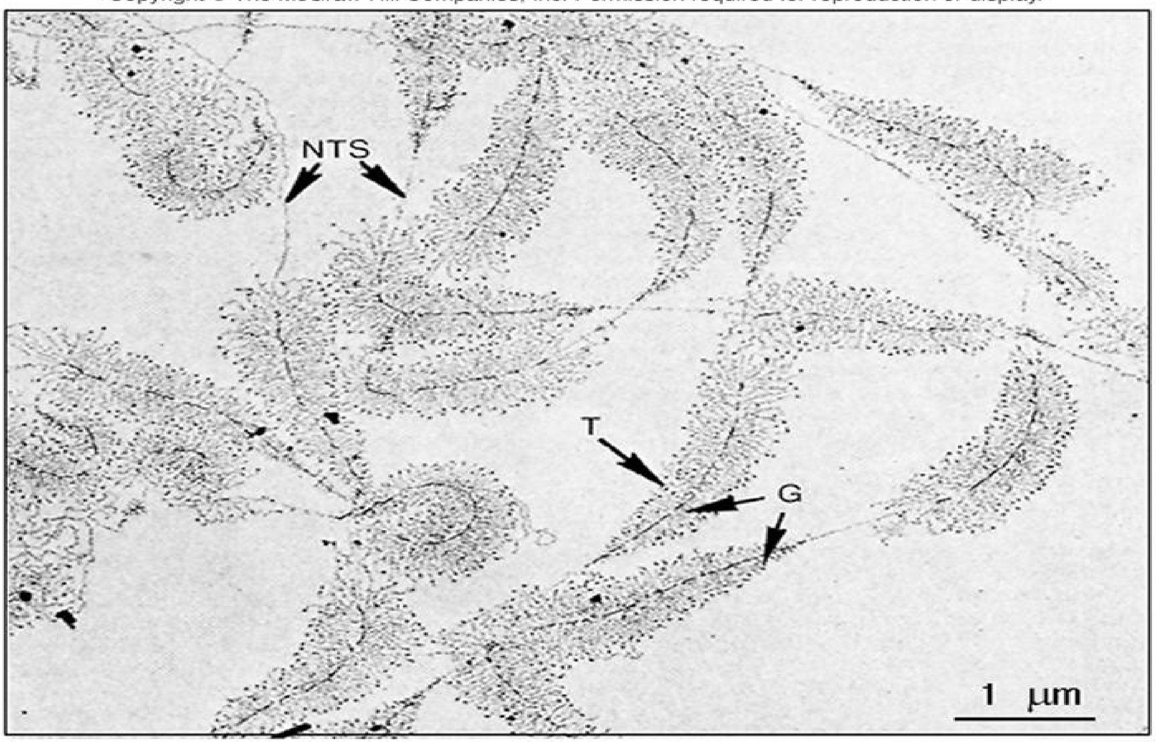
rRNA Processing

1) Eukaryotic rRNA Processing:
(snoRNAs and snoRNPs)

2) Prokaryotic rRNA Processing:
(RNase III and RNase E)

Transcription of rRNA precursor genes





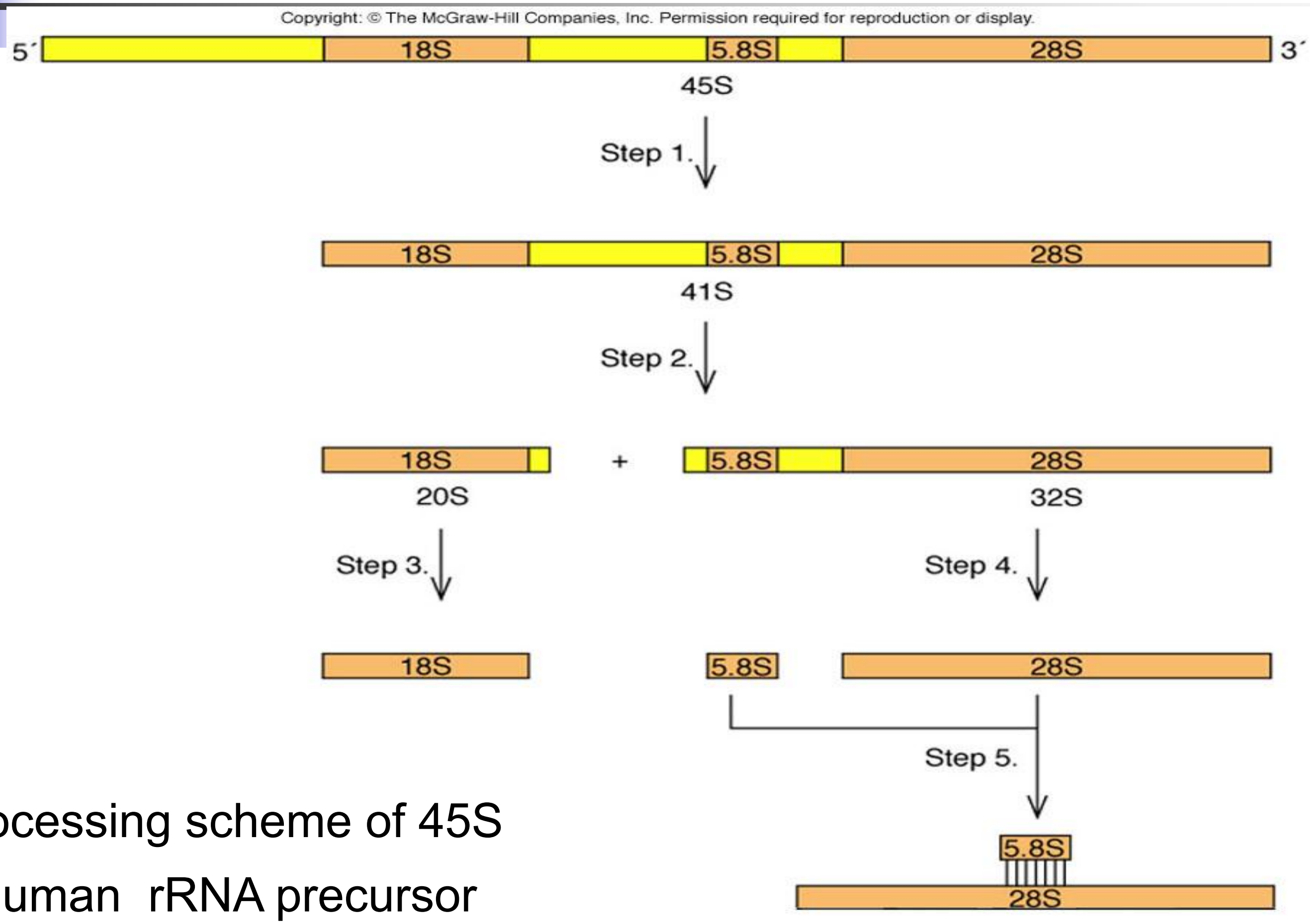
(Source: O.L. Miller, Jr., B.R. Beatty, B.A. Hamkalo, and C.A. Thomas, Electron microscopic visualization of transcription. *Cold Spring Harbor Symposia on Quantitative Biology* 35 (1970) p. 506.)

rRNA are separated by regions called nontranscribed spacers (NTS). NTS are distinguished from transcribed spacers. They will be removed in the processing of the precursor to mature rRNA species.

NTS**:** nontranscribed spacer

T: rRNA precursor transcripts
G: duplicated rRNA precursor genes

Eukaryotic rRNA Processing



Processing scheme of 45S human rRNA precursor

Something involved:

Small nucleolar RNAs (snoRNAs)

Small nucleolar ribonucleoprotein (snoRNPs)

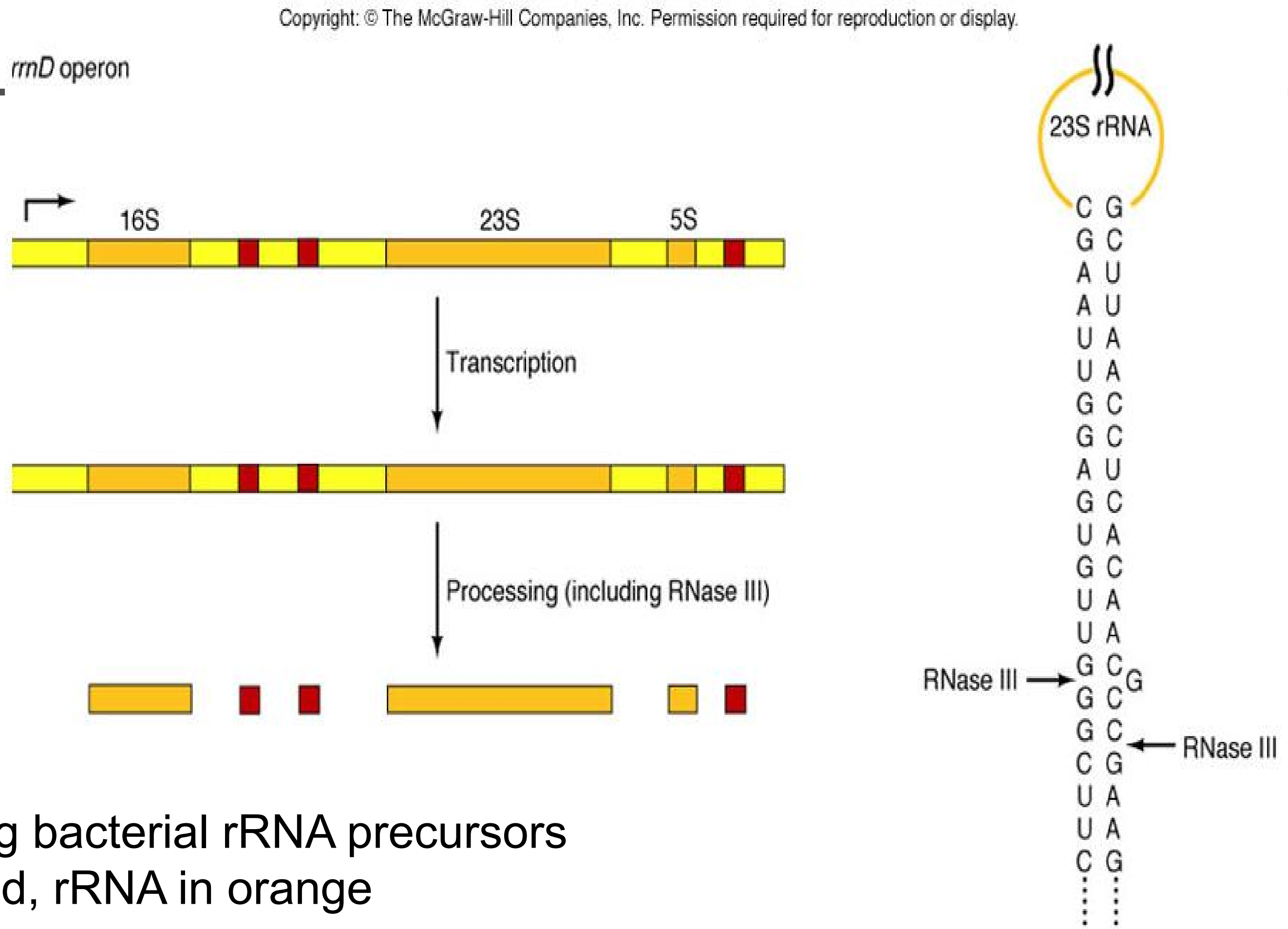
E1, E2 (snoRNPs) : 18S rRNA maturation

E3 (snoRNP) : Cleavage upstream of 5.8S rRNA

Summary

- rRNAs are made in eukaryotic nucleoli as precursors that must be processed to release the mature rRNAs.
- The order of RNAs in the precursor is 18S, 5.8S, 28S in all eukaryotes, although the exact sizes of the mature rRNAs vary from one species to another.
- snoRNAs play vital roles in these processing steps.

Prokaryotic rRNA Processing:



Processing bacterial rRNA precursors
tRNA in red, rRNA in orange

Prokaryotic rRNA precursors contain tRNAs as well as all three rRNAs. The rRNAs are released from their precursors by RNase III and RNase E

tRNA Processing

1. Cutting apart polycistronic (多顺反子) precursors

    (RNAse III)

2. Forming mature 5’ –ends

    (RNAse P)

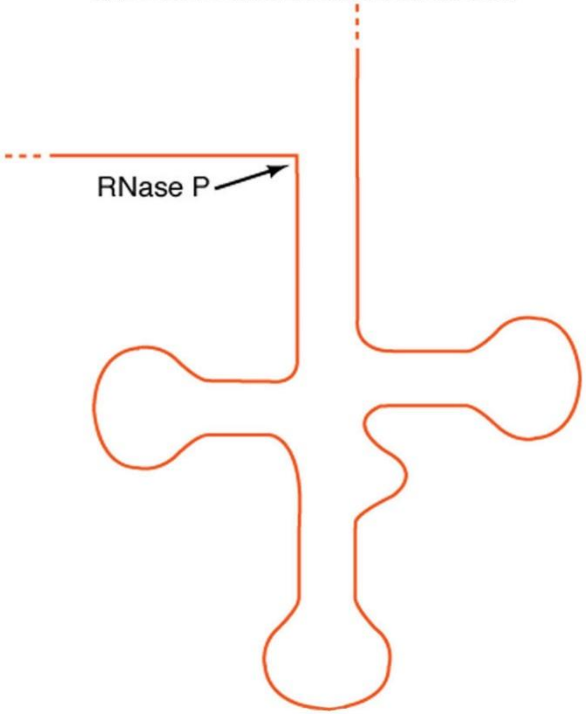
3. Forming mature 3’ –ends

Forming mature 5' – ends

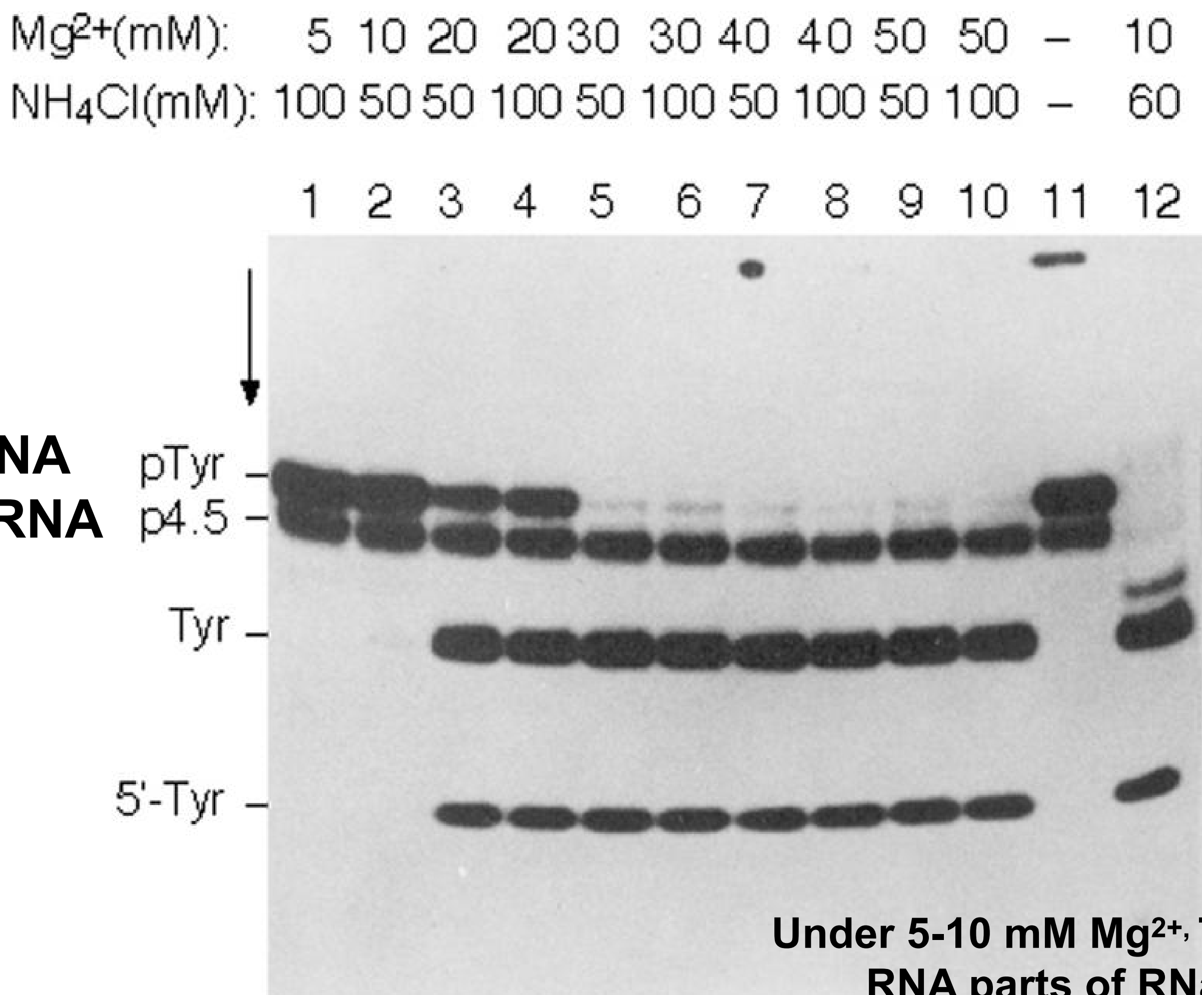
RNAse P: contains two subunits. one of these subunits is made of RNA , not protein.

M1RNA:125 kD

Protein: 14 kD



M1RNA of *E.coli* RNAse P has enzymatic activity



Crude RNaseP

Lane 1-11 purified M1RNA

Pre-tRNA
P4.5s RNA

Under 5-10 mM Mg^{2+}, The protein and the RNA parts of RNaseP are required

(Source: Guerrier-Takada C., K. Gardiner, T. Marsh, N. Pace, and S. Altman, The RNA moiety of ribonuclease P is the catalytic subunit of the enzyme. *Cell* 35 (Dec 1983) p. 851, f. 4A. Reprinted by permission of Elsevier Science.)

Forming mature 3’ –ends

RNAse D

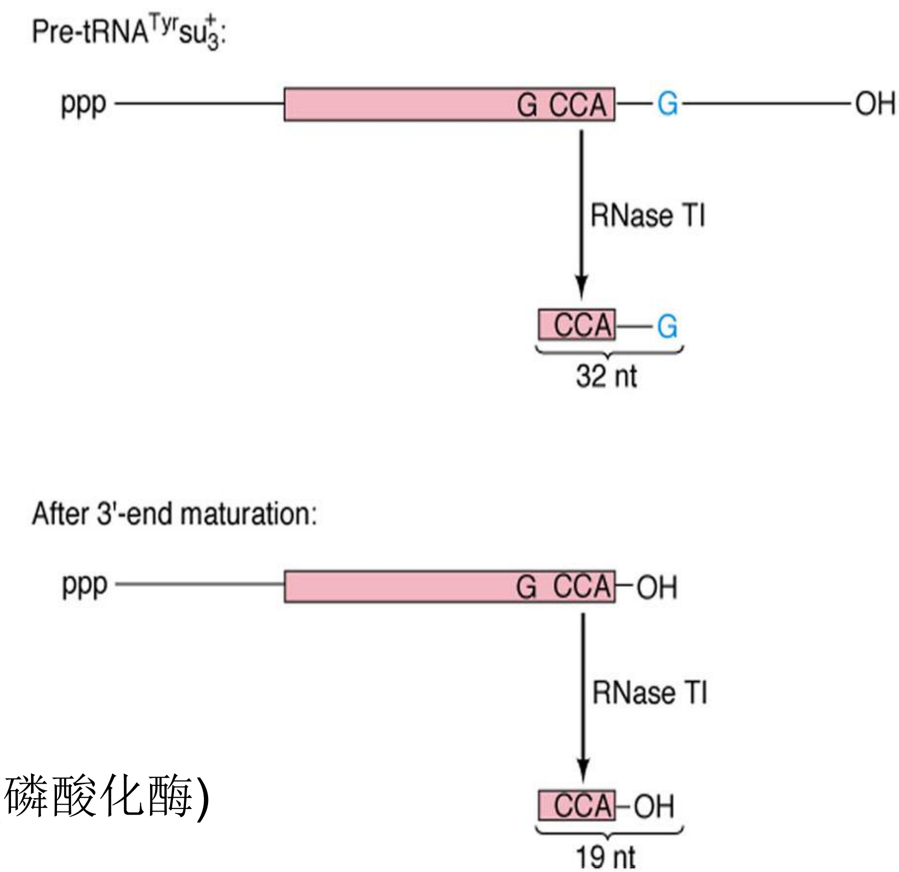
RNAse BN

RNAse T

RNAse PH

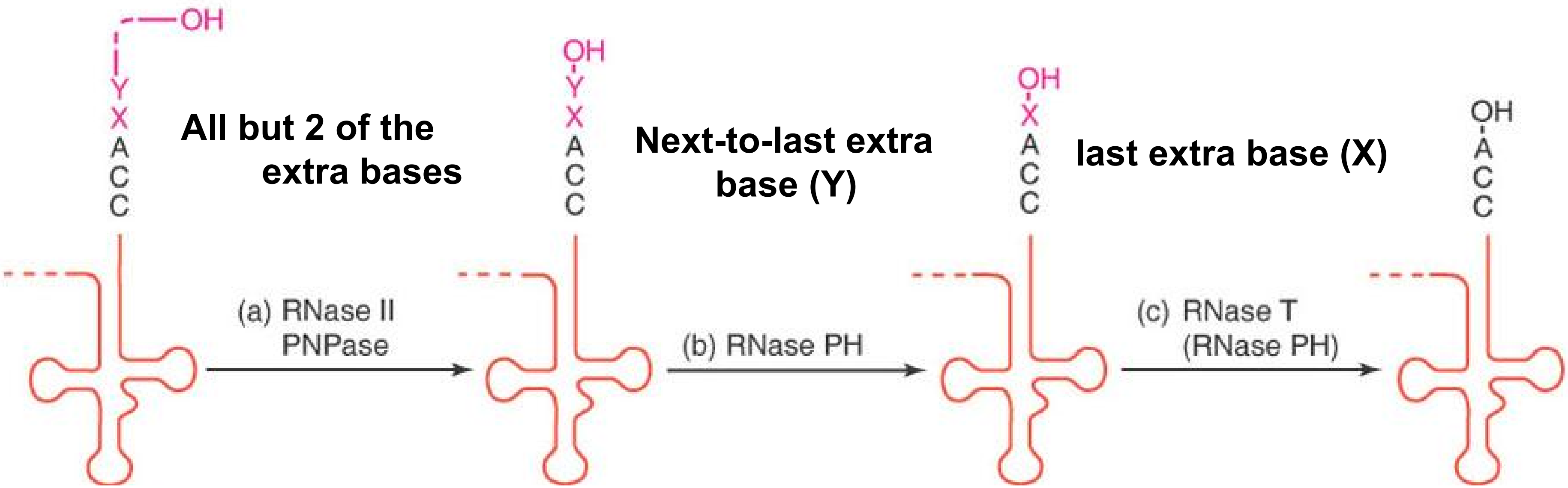
RNAse II

Polynucleotide phosphorylase (磷酸化酶)
(PNPase)



Substrate for in vitro assay of tRNA 3 ’ –end maturation

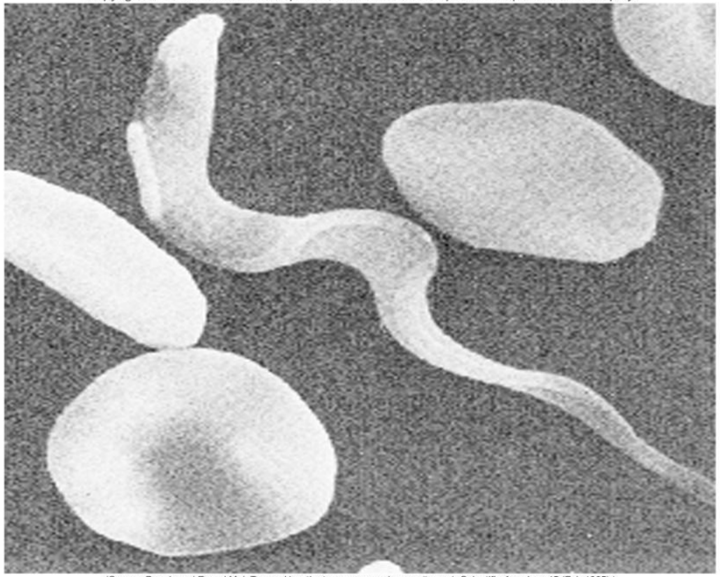
Model for processing the 3'-end of an *E.coli* tRNA



Trans-Splicing

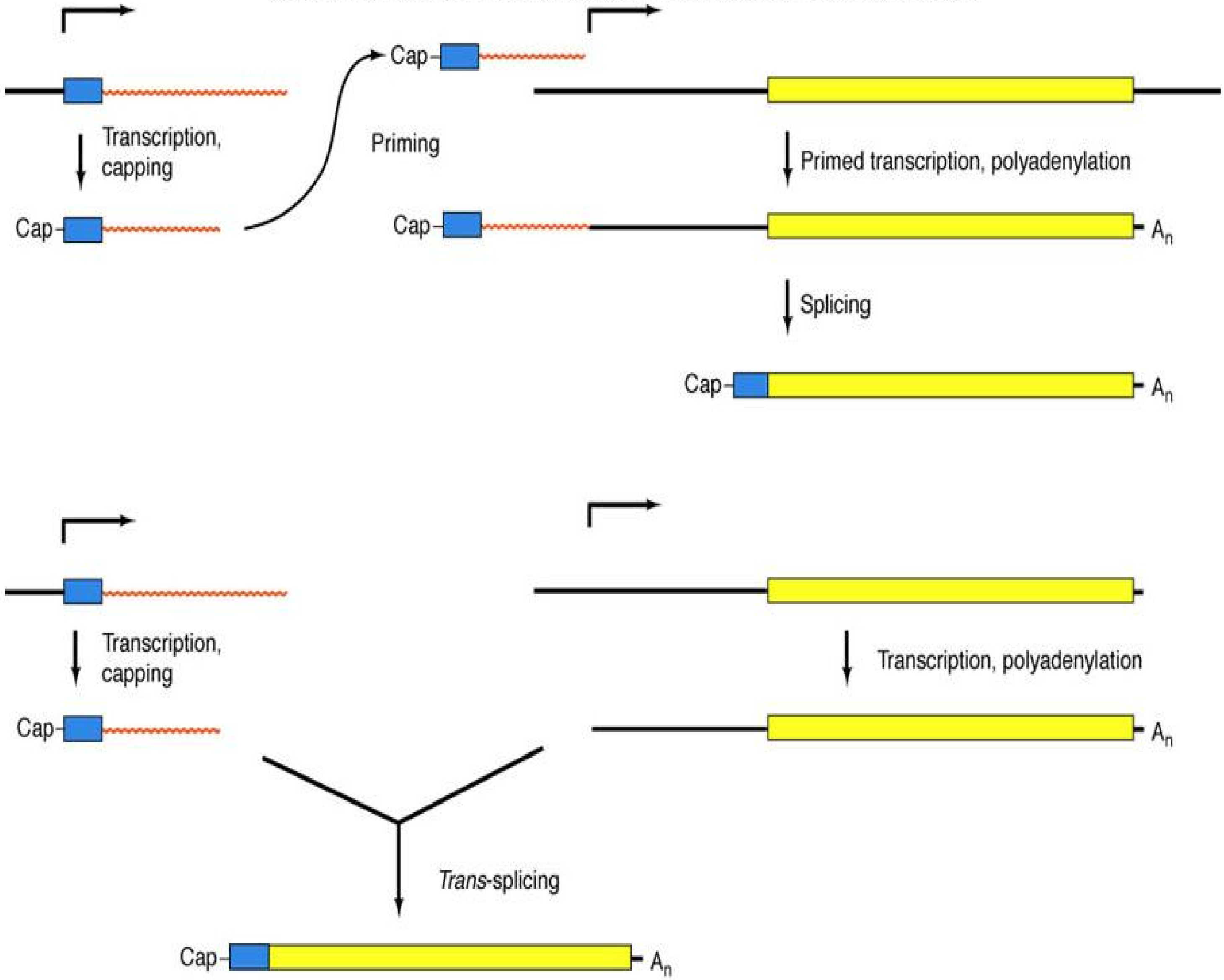
The splicing we talked previously can be called cis-splicing **because it involves 2 or more exons that exists together in the same gene.**

Trans-splicing, the exons are not part of the same gene at all and may not even be found on the same chromosome.



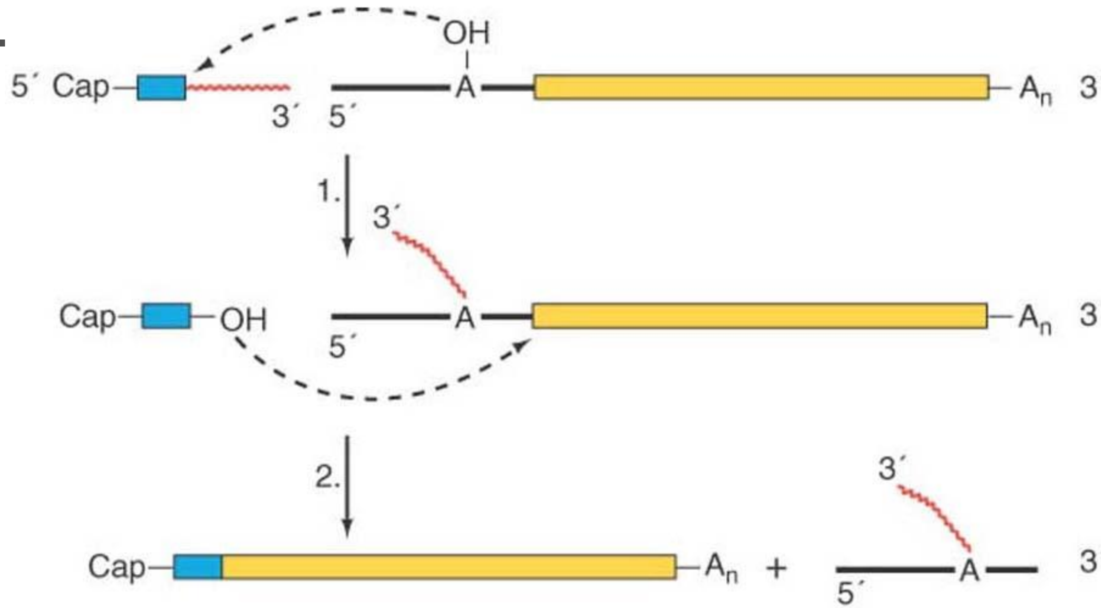
(Source: Donelson J.E. and M.J. Turner, How the trypanosome changes its coat. *Scientific American* 45 (Feb 1985).)

Trypanosome (锥体虫)



Two hypotheses for join the SL (spliced leader) to the coding region of an mRNA

Detailed trans-splicing scheme for a trypanosome mRNA



Step1: the branchpoint A within the half-intron attacks the junction between the leader exon and its half-intron. A Y-shaped intron-exon intermediate formed

Step2: the leader exon attacks the splice site between the branched intron and the coding exon. Produced the spliced, mature mRNA+ the Y-shaped intron.

RNA editing

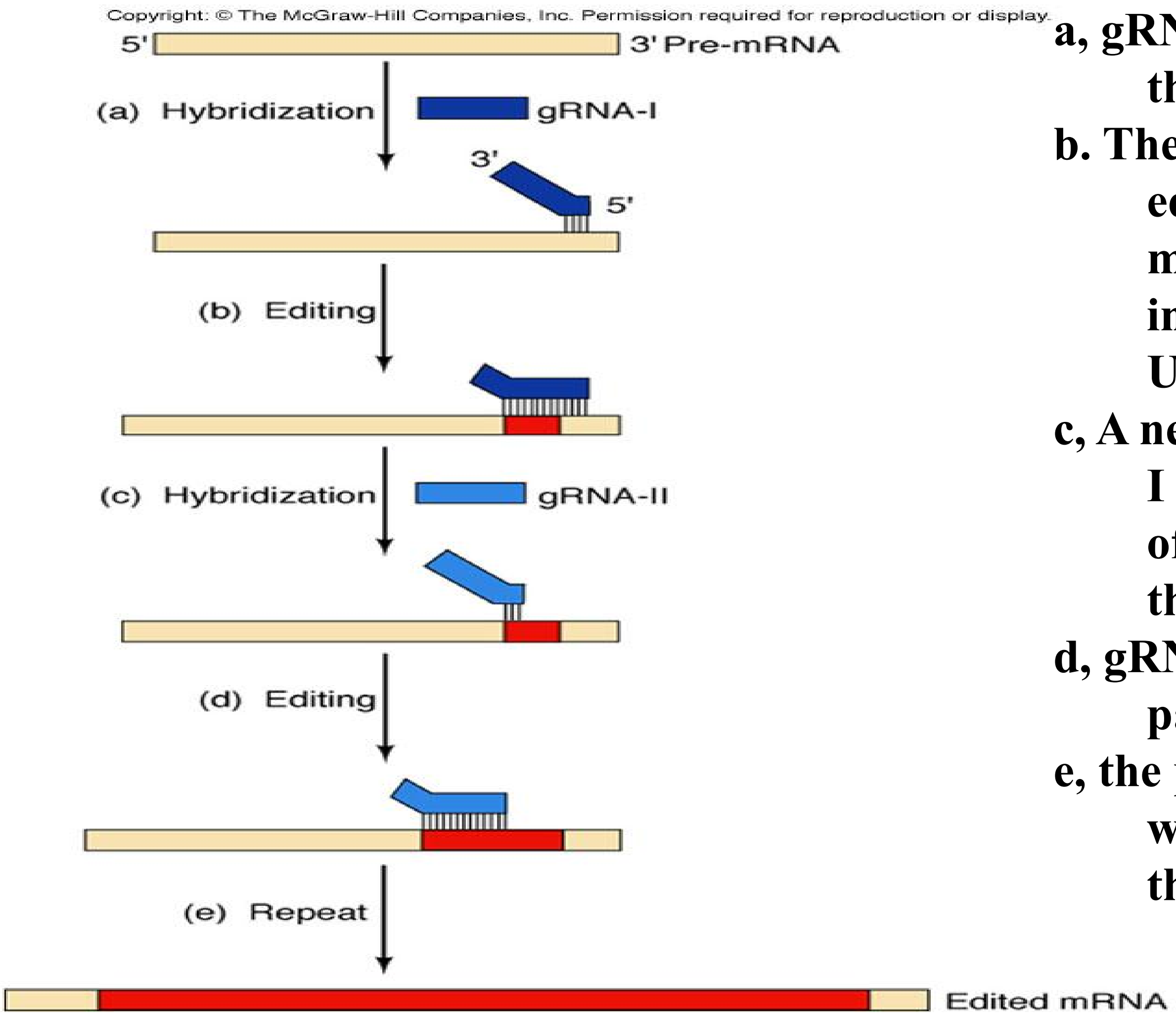
In trypanosome

Some Us are added or deleted in the mature mRNA during the maturation, this is called RNA editing

COX II DNA: ···GTATAAAAGTAGA   G A ACCTGG···

COX II RNA: ···GUAUAAAAGUAGAUUGUAUACCUGG···

RNA editing



a, gRNA hybridizes to a region of the pre-mRNA.

b. The rest of the gRNA-I directs editing of part of the pre-mRNA. The pre-mRNA grow in length due to the inserted UMPs.

c, A new gRNA-II displaces gRNA-I by hybridizing to the 5'-end of the newly edited region of the pre-mRNA

d, gRNA-II directs editing of a new part of the pre-mRNA

e, the previous steps are repeated with additional gRNAs until the RNA is completely edited.

Model for the role of gRNAs in editing

Student’s presentation (today)

Paper: Zhong et al. Cysteine-rich peptides promote interspecific genetic isolation in Arabidopsis. Science, 2019, 364, DOI: 10.1126/science.aau9564

Presenter: Group 9